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foundationally, adversarially and from first-principles attack this whitepa

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foundationally, adversarially and from first-principles attack this whitepaper I'm busy writing. the intent should be clear.

Attack the mathematics AND the physics. I'm both a

Show more

I'll treat this as a hostile technical referee report: isolate the core claim chain, separate proved/imported/open material, and attack definitions, normalizations, hidden assumptions, and overclaims.

The main vulnerability is sharper now: the determinant/K_0 package may be clean but tautological, while the physics language risks upgrading a recognition ledger into realization.

Thought for 11m 9s

I'm going to be brutal. The draft is strongest as a **normalization-and-separation paper**: it carefully distinguishes the Borcherds product, the Gritsenko–Nikulin denominator algebra, the OP/DT scalar square, and an open compact Hall/factorization realization. It is weakest whenever the prose makes the reader feel that these layers are closer than they are. The main danger is not that the automorphic mathematics is fake; the main danger is that the physical and Hall-theoretic interpretation is not yet mathematically supported by an additive charge category.

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Executive verdict

The paper's core theorem is basically:

$$\phi_{0,1} \quad \text{Borcherds/Gritsenko–Nikulin} \quad \Delta_5$$

with a careful convention that the theta-normalized form has leading coefficient 64, so that the monic product is $D_5 = 64^{-1}\Delta_5$. That part is fine and externally anchored: Gritsenko–Nikulin explicitly write $64^{-1}\Delta_5$ as the product with exponents $f(nm, l)$, the Fourier coefficients of $\phi_{0,1}$. arXiv +1

But the paper risks sounding like it has built a **BPS square root** for $K3 \times E$. It has not. It has built a **virtual K_0-determinant** and aligned it with an existing BKM denominator theorem. Your own draft repeatedly admits this: the half-index object is only a $K_0(\mathrm{sVect})$ representative with no canonical Hilbert-space half, no parity lift, and no operator product; the compact Hall/chiral realization is explicitly open.

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The single most serious foundational problem is this:

$$(n, l, m)$$

is being used in three incompatible ways: Fourier exponent, Lorentzian BKM root degree, and would-be physical/Hall charge degree. The first two are additive. The third is generally **not** additive if (n, l, m) are Gram invariants of physical charges.

That has to be fixed before any compact Hall/factorization story can be more than a recognition wish-list.

1. Fatal issue: Gram invariants are not physical Hall charges

You define

$$\Gamma_{\text{gram}} = \Gamma_{\text{ind}} = \mathbb{Z}^3$$

and explicitly say it is "notation for Gram invariants and Fourier indices," not a microscopic charge lattice. Then later the compact Hall discussion assumes an extension-closed Hall category with numerical Gram-index lattice Γ_{ind}, semistable stacks indexed by $\gamma \in \Gamma_{\text{ind}}$, and Hall correspondences landing in $\gamma + \eta$.

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That is not a small technical gap. It is structurally wrong unless you supply a new construction.

If physical charges are electric-magnetic pairs (Q, P), then

$$n = \frac{Q^2}{2}, \qquad l = Q \cdot P, \qquad m = \frac{P^2}{2}.$$

But for two charge pairs,

$$\frac{(Q_1 + Q_2)^2}{2} = \frac{Q_1^2}{2} + \frac{Q_2^2}{2} + Q_1 \cdot Q_2,$$

$$(Q_1 + Q_2) \cdot (P_1 + P_2) = Q_1 \cdot P_1 + Q_2 \cdot P_2 + Q_1 \cdot P_2 + Q_2 \cdot P_1,$$

$$\frac{(P_1 + P_2)^2}{2} = \frac{P_1^2}{2} + \frac{P_2^2}{2} + P_1 \cdot P_2.$$

So the map

$$(Q, P) \mapsto (n, l, m)$$

is quadratic. It does **not** send physical charge addition to addition in $\mathbb{Z}^3$, except in special cross-orthogonal situations. A Hall algebra must be graded by an additive K-theory, Mukai, or electromagnetic charge lattice. It cannot be fundamentally graded by the Gram invariants unless you prove a quotient/descent theorem saying that the Hall product descends through this nonlinear invariant map.

The same issue appears on the D6–D2–D0 side. You define rank-one charges

$$\gamma(h, d, n) = (1; (\beta_h, d); n), \qquad \beta_h = s_1 + hF,$$

and the scalar DT series sums over h, d, n. But the rank-one D6 sector is not extension-closed under Hall multiplication: extensions/direct sums of rank-one objects naturally leave the rank-one sector. Also $h - 1 = \beta^2/2$ is quadratic in β, not additive. The family $\beta_h = s_1 + hF$ is not closed under addition in the way a Hall grading would require.

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What a hostile referee will say

"You have a beautiful additive root lattice for the automorphic product, but you have not shown that it is the additive charge lattice of the compact BPS category. You have shown an additive algebra of Fourier exponents, not an additive algebra of physical charges."

Required repair

Introduce two lattices, not one:

$$\Gamma_{\text{phys}} \quad \text{and} \quad \Gamma_{\text{gram}}.$$

The Hall category must be graded by Γ_{phys}. Then define a non-additive invariant map

$$\pi : \Gamma_{\text{phys}} \to \Gamma_{\text{gram}}.$$

The automorphic product is the π-pushed-forward index. If you want a primitive Lie algebra graded by Γ_{gram}, you must prove a descent theorem:

$$\mathrm{Prim}_{\Gamma_{\text{phys}}} \longrightarrow \mathrm{Prim}_{\Gamma_{\text{gram}}}$$

compatible with brackets. Without this, the compact Hall/factorization target cannot literally be $\mathfrak{g}_{\Delta_5}$-graded by (n, l, m).

2. The “half K3 index” is not physical data

The draft says:

$$Z_{K3} = 2\phi_{0,1}, \qquad \phi_{0,1} = \sum f(n,l) q^n r^l.$$

Then it introduces

$$\mathcal{U}_{n,l} \in K_0(\mathrm{sVect}_{fd}), \qquad \mathrm{sdim}\, \mathcal{U}_{n,l} = f(n,l),$$

and correctly says this is not a canonical half of the K3 BPS Hilbert space.

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That is honest, but it also means the phrase “one-particle index” is dangerous. A hostile physicist will object:

$$\mathrm{sdim}\, H^{\mathrm{BPS}}_{n,l} = 2f(n,l)$$

is an index of the K3 sigma model. The half-index $f(n,l)$ is an arithmetic input to the Borcherds lift. It is not a protected subspace, not a superselection sector, and not a Hilbert-space trace.

The situation is even sharper for negative coefficients. If $f(n,l) < 0$, your model takes an odd vector space of dimension $-f(n,l)$. That is a perfectly valid K_0-representative.

But it is arbitrary. Many parity splits have the same superdimension. The determinant only sees

$$\dim V_0 - \dim V_1,$$

not $(\dim V_0, \dim V_1)$. Your own Proposition 10.29/10.30 says the denominator product does not determine parity dimensions, zero-superdimension spaces, or structure constants. raeez.lorgat.automorphic-correc...

Required repair

Stop calling $\mathcal{U}_{n,l}$ a "half-index object" without immediately saying:

> This is an arbitrary Grothendieck representative of the arithmetic half-index. It has no physical Hilbert-space status.

Rename

$$Z^{\mathrm{vir}}_{\mathrm{BPS}}$$

to something like

$$Z^{\mathrm{vir}}_{\mathrm{Borch}}$$

unless and until a compact BPS realization is actually constructed.

3. The main determinant theorem is mostly an imported product theorem

The paper's determinant proof is:

1. Choose a K_0-graded virtual object with superdimensions $f(nm, l)$.
2. Apply the formal Fock/superdeterminant identity.
3. Invoke the Gritsenko–Nikulin Borcherds product to identify the result with Δ_5.

This is mathematically valid, but it is not a new construction of Δ_5, nor of a BPS determinant. Gritsenko–Nikulin already give the product

$$\frac{1}{64}\Delta_5(Z) = e^{\pi i(z_1+z_2+z_3)} \prod_{(n,l,m)>0} \left(1 - e^{2\pi i(nz_1+lz_2+mz_3)}\right)^{f(nm,l)}.$$

That is exactly the spine of your determinant. arXiv +1

So the contribution is not “we derive the Igusa square root.” The contribution is:

> We package the known Borcherds product as a virtual K_0-superdeterminant, fix the theta-vs-monic normalization, and separate it from the OP scalar square and the open compact realization problem.

That is a respectable paper. It is just a more modest one.

Required repair

Change theorem names like “Virtual protected determinant” and “main synthesis theorem” so that they do not sound like new physical construction. Use titles like:

> Normalized K_0-determinant form of the Gritsenko–Nikulin product.

4. The OP/DT scalar square is a scalar theorem, not evidence for the square root

The OP/DT part is externally legitimate: Oberdieck–Pixton prove the primitive $K3 \times E$ Igusa cusp form conjecture in the reduced Gromov–Witten setting, and Oberdieck’s reduced stable-pair work compares quotient Behrend-weighted definitions and reduced virtual-class definitions. arXiv +1

But the physics/mathematics relation is easy to overstate. Bryan’s account makes the scalar conjecture

$$DT(X) = -\frac{1}{\chi_{10}}$$

with the monic product convention for χ_{10}, and also emphasizes that ordinary DT invariants vanish because of the E-action, forcing a reduced quotient theory.

UBC Mathematics

Your conversion

$$\chi_{10}^{OP} = D_5^2 = (64^{-1}\Delta_5)^2, \qquad Z_{OP}^X = -4096\Delta_5^{-2}$$

is fine as a normalization statement. But it has no physical content beyond normalization. The number 4096 is not a degeneracy, not an anomaly, not an orientation character, and not a square-root datum. You say this, but the scalar-square section is still positioned so prominently that readers may infer more.

Required repair

Move the scalar square after the denominator algebra, not before the physical interpretation. Present it as:

> The OP theorem sees D_5^2, not D_5. Therefore it confirms the scalar square only in the reduced enumerative theory. It supplies no square-root orientation.

5. Your “primitive recognition theorem” is nearly tautological

The compact recognition theorem assumes:

- oriented Hall object,
- protected integration,
- chiral bar-cobar comparison,
- primitive representatives,
- parity split,
- Chevalley relations,
- real Serre relations,
- imaginary orthogonality relations,
- no-extra-relations theorem,
- completed PBW compatibility,
- same radical quotient,
- full root-space parity dimensions.

Then it concludes:

$$H^0\mathrm{Prim}^{prot}(F_X) \cong \mathfrak{g}_{\Delta_5}.$$

This is formally true, but a hostile reader will call it a restatement of the desired isomorphism. Your Definition 10.19 already packages essentially everything required for the conclusion. raeez.lorgat.automorphic-correc...

The theorem is useful as a **certificate checklist**. It is weak as a theorem.

Required repair

Rename it:

> Recognition certificate for a compact Igusa realization.

Then state:

> This is not a theorem producing new structure. It is a list of data whose verification would be sufficient to identify a compact primitive algebra with $\mathfrak{g}_{\Delta_5}$.

That will make it harder to attack.

6. The E_3-factorization language is underdefined

You repeatedly refer to a "holomorphic E_3-factorization algebra" on the Calabi–Yau threefold $K3 \times E$. You also say the symbol is a placeholder and not a constructed functor. raeez.lorgat.automorphic-correc…

That is good honesty, but the phrase itself is not standard enough to carry the argument. In a real six-dimensional topological theory one expects E_6-type local structure. In a holomorphic theory on a complex threefold one gets a holomorphic factorization algebra, but not automatically an ordinary little-disks E_3-algebra unless a precise holomorphic/formality/framing reduction is supplied. You list "formality point" and "S^3-framing witness," but this is currently a slogan, not a definition.

Hostile objection

"What exactly is an E_3-algebra on a complex threefold here? Is this holomorphic disks, real little 3-disks, Costello–Gwilliam factorization, E_3 from complex dimension three, or a framed E_3 after integrating over K3? Which operad acts, and on what category?"

Required repair

Define the exact structure before using the symbol:

$$A_X \in \mathrm{Alg}_{\mathcal{O}}(\mathrm{Fact}_{hol}(X))$$

for some named operad or factorization category. If "E_3" is only heuristic, remove it from theorem-level statements and say "holomorphic factorization algebra on X" until the operadic structure is specified.

7. The $K3 \to E$ specialization is the real missing construction

The paper wants

$$F_X \simeq \mathrm{Sp}^{ch}_{K3,E}(A_X^{E_3}) \in \mathrm{Fact}(\mathrm{Ran}(E)).$$

But $\mathrm{Sp}^{ch}_{K3,E}$ is not a known functor in the paper. It is a desired operation. You admit that construction is open. raeez.lorgat.automorphic-correc...

The attack is simple: all compact physics is hidden inside the undefined functor.

A reader can accept the Borcherds product without accepting any of the chiral/factorization narrative. The paper should therefore not let the undefined specialization appear as if it explains the product.

Required repair

Make the logical direction one-way:

$$\text{Known automorphic denominator} \quad \Rightarrow \quad \text{target constraints for any future } K3 \to E$$

Do not suggest that the specialization is part of the proof.

8. The Ran(E) obstruction is good, but phrase it narrowly

Your observation is correct: if a one-dimensional subscheme $C \subset K3 \times E$ has positive elliptic degree, then $p_E(C) = E$. Therefore such a curve cannot be localized at finitely many points of E. The manuscript uses this to argue that naive support-local factorization on $\mathrm{Ran}(E)$ sees only projection-finite sectors.

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That is one of the strongest original conceptual points in the draft.

But do not overstate it as a general obstruction to locality. It is an obstruction to a **particular pushforward-to-Ran(E)** model. A six-dimensional local theory on $K3 \times E$ can still have local observables; the problem is that positive elliptic-degree curve sectors are global from the E-projection viewpoint.

Required repair

Replace “locality obstruction” with:

projection-to-E support-locality obstruction.

That avoids a physicist objecting that you are confusing spacetime locality with support-locality after dimensional reduction.

9. The BKM denominator is not a microscopic BPS algebra

The denominator algebra is real mathematics: Gritsenko–Nikulin construct a generalized Lorentzian Kac–Moody superalgebra whose denominator is the normalized Δ_5 form. Their paper explicitly gives Δ_5 as the correcting automorphic form for the rank-three Cartan matrix with entries 2 on the diagonal and -2 off the diagonal. arXiv

But the BKM algebra is not automatically the algebra of BPS states. You already know this and state it. The determinant only fixes signed supermultiplicities:

$$\mathrm{smult}(\alpha) = \dim \mathfrak{g}_{\alpha,0} - \dim \mathfrak{g}_{\alpha,1}.$$

It does not fix the two dimensions separately, invisible zero-superdimension root spaces, or brackets. raeez.lorgat.automorphic-correc...

A very simple adversarial counterexample is already implicit in your paper:

Take any graded super vector space B with

$$\mathrm{sdim}\, B_{n,l,m} = f(nm, l)$$

and give it the zero bracket. It has the same determinant, but it is not $\mathfrak{g}_{\Delta_5}$. Your own Proposition 10.31 says the determinant forgets brackets, parity splits, simple-root representatives, relation constants, and Hopf-pairing radicals.

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Required repair

Every time you write

$$\mathrm{smult}\, \mathfrak{g}_{\Delta_5,\alpha(n,l,m)} = f(nm, l),$$

immediately distinguish:

$$\text{signed root Euler characteristic} \neq \text{state space} \neq \text{primitive Hall bracket}.$$

You do this in places. Do it everywhere.

10. The physical duality group is underspecified

The paper uses $Sp_4(\mathbb{Z})$ as the automorphic group of Δ_5 and then discusses physical charge invariants. That is fine automorphically, but physically $K3 \times E$ type II compactification has a much larger charge/duality structure. $Sp_4(\mathbb{Z})$ acts naturally on

genus-two chemical potentials and the triple of T-duality invariants, not on the full microscopic charge lattice.

So the paper needs to say:

$$Sp_4(\mathbb{Z})$$

is the modular group of the genus-two counting function / invariant chemical potential, not the full physical U-duality group of the compactification.

Otherwise a physicist will object that the true BPS charge lattice and wall-crossing data contain more information than $(Q^2/2, Q \cdot P, P^2/2)$. Multi-centered wall crossing, for example, depends on splittings and symplectic pairings of constituents, not only the final Gram matrix.

Required repair

Add a paragraph:

> The $Sp_4(\mathbb{Z})$ action here is the arithmetic symmetry of the genus-two automorphic form and of the reduced invariant variables. It is not asserted to be the full U-duality action on the microscopic charge lattice.

11. The $m = 0$ boundary factors are not one-particle K3 states

You define

$$V_{n,l,m} = \mathcal{U}_{nm,l}.$$

For $m > 0$, this matches the Borcherds/DMVV-style exponent $f(nm, l)$. But for $m = 0$, you get infinitely many boundary factors with the same K3 coefficient $f(0, l)$ as n varies. Those are cusp/Weyl/oscillator boundary factors in the product. They are not literal copies of a K3 BPS Hilbert space indexed by arbitrary n.

You note that $m = 0$ are cusp-boundary terms, but the “one-particle object” language still invites a wrong physical reading.

Required repair

Split the virtual object:

$$V = V_{\text{bulk}} \oplus V_{\text{cusp}}.$$

Then state:

- V_{bulk} is the Borcherds/Hecke exponent package for $m > 0$;
- V_{cusp} is a boundary/Weyl correction package;
- neither is a microscopic one-particle Hilbert space.

12. Boundary rows: good ledger, but too much row-map danger

The Cléry–Gritsenko result really is an eight-row classification of genus-two Siegel modular forms with the simplest diagonal divisor. arXiv

Govindarajan's CHL paper also states that the square roots of the relevant twisted dyon Siegel forms are precisely the dd-modular forms classified by Cléry–Gritsenko and arise as Weyl–Kac–Borcherds denominators. arXiv

So your ledger is defensible. But this part is high-risk because you are mixing:

- Cléry–Gritsenko row order,
- your row order,
- CHL $G_N(M)$ notation,
- product-degree lattices,
- multiplier covers,
- scalar squares,
- denominator identities,
- physical hosts.

A hostile reader will search for one row-map mismatch and use it to discredit the appendix.

Required repair

Move the eight-row discussion to a separate paper or a clearly optional appendix. If kept, every row needs a compact table with:

$$(F_j, \Gamma_j, \mathrm{cover}, \nu_j, \mathrm{seed}, \rho_j, L_j, \mathrm{source\ theorem}).$$

For $F_7 = \nabla_{3/2}$, do not merely say "μ_4-cover." Explicitly state which multiplier is pulled back, what becomes single-valued, and which square-root branch is used.

13. The L-function appendix is a distraction

The spin L-factor normalization for Δ_5^2 is not needed for the main paper. Even if correct, it invites a referee to audit a completely separate Saito–Kurokawa normalization chain.

Your main argument does not require:

$$L_{\mathrm{spin}}(s, \Delta_{10}) = \zeta(s-8)\zeta(s-9)L(s, \Delta E_6).$$

It does not help construct the square root, the Hall orientation, the compact factorization algebra, or the primitive bracket. It is a liability.

Required repair

Remove it, or move it to a separate note titled “normalization appendix, independent of the main theorem.”

14. The current-envelope construction is formal, not geometric

You define

$$\mathrm{Cur}_E(A_{\mathrm{den}}) = A_{\mathrm{den}} \otimes \mathcal{O}_E$$

and then

$$\mathrm{Den}_{\Delta_5,E} = U_E^{ch}(\mathrm{Cur}_E(A_{\mathrm{den}})).$$

This is fine as a formal Beilinson–Drinfeld current algebra. But it is not a compact BPS chiral algebra. The construction works for any Lie superalgebra; it does not use $K3 \times E$ geometry except by naming E.

Your paper says this, but the notation $\mathrm{Den}_{\Delta_5,E}$ may still make readers think the curve E has acquired physical content.

Required repair

Call it:

$$\mathrm{CurEnvelope}_E(\mathfrak{g}_{\Delta_5})$$

and say:

> This is a formal target object, not a result of compactification.

15. The strongest version of the paper is much simpler

I would restructure the manuscript around a cleaner thesis:

Theorem A: normalization

$$D_{\phi_{0,1}}(Z) = 64 q^{1/2} r^{1/2} s^{1/2} \prod_{\Gamma_{\text{eff}}} (1 - q^n r^l s^m)^{f(nm,l)} = \Delta_5(Z).$$

This is the theta-normalized Gritsenko–Nikulin product.

Theorem B: denominator target

$$\operatorname{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1} \Delta_5(2Z), \qquad \operatorname{smult} \mathfrak{g}_{\alpha(n,l,m)} = f(nm, l)$$

on active support.

Theorem C: scalar square normalization

$$Z^X_{OP/DT} = -4096 \Delta_5^{-2} = -(64)^2 D_{\phi_{0,1}}^{-2}.$$

Open Problem D: compact realization

Construct a real additive physical charge lattice, Hall category, orientation, protected integration, chiral specialization, and primitive bracket whose invariant pushforward recovers the BKM denominator.

This framing would be almost unassailable.

Bottom-line attack

The paper is mathematically credible where it says:

$$\text{Borcherds product of } \phi_{0,1} = \Delta_5, \qquad \text{GN denominator } = 64^{-1} \Delta_5(2Z), \qquad \text{OP scal}$$

It becomes vulnerable where it lets the reader believe:

$$\Delta_5^{-1}$$

is a constructed compact BPS partition function, or that

$$(n, l, m)$$

is already a physical Hall charge. It is not. It is a Fourier/root degree. The physical charge lattice problem is the central unresolved issue.

Your best defense is to lean into that: make the paper a precise **no-overclaim ledger**. The title could remain provocative, but the theorem statements should be austere. The compact $K3 \times E$ realization should be framed as a recognition problem constrained by the automorphic data, not as something the determinant has substantially built.

Sources

Sources

No you don't seem to get it.

The point is NOT to cut the paper down to size. it's to understand what it is trying to prove and produce those constructions, derivations, deductions & mathematics

Show more

Let's construct, deduct, prove the The physical charge lattice problem which is the central unresolved issue.

You're right. I'll switch from referee containment to constructive reconstruction: identify the intended compact Igusa realization and build the missing mathematical and physical machinery as far as possible.

The first construction should be a two-layer charge formalism: additive microscopic charges upstairs, additive Gram-current degrees downstairs, linked by a non-additive moment map rather than quotienting physics away.

I'll aim to prove recovery of the Igusa/BKM $\mathbb{Z}^3$-grading as an invariant pushforward or sector degree, with convolution induced by block-sum/orthogonal decompositions.

Thought for 11m 24s

Yes. Let's build the missing object.

The correct resolution is:

$$\Gamma_{\text{phys}} \neq \Gamma_{\text{ind}}.$$

The physical Hall/brane charge lattice is additive. The Igusa/Borcherds lattice $\Gamma_{\text{ind}} \cong \mathbb{Z}^3$ is the **rank-two Gram invariant** of that additive lattice. The construction is not to force (n, l, m) to be microscopic; it is to prove that (n, l, m) is the invariant shadow of a real additive physical charge lattice, and then define the correct descent.

Your paper already has the ingredients: the D6–D2–D0 branch, the OP variables, the Gram convention, and the statement that Γ_{ind} is not itself microscopic. The missing construction is the **dyonic Mukai lattice over $K3 \times E$.** raeez.lorgat.automorphic-correc...

1. The physical charge lattice

Let S be the K3 surface and $X = S \times E$. Put

$$\Lambda_S := H(S, \mathbb{Z}) = H^0(S, \mathbb{Z}) \oplus H^2(S, \mathbb{Z}) \oplus H^4(S, \mathbb{Z})$$

with Mukai pairing

$$(r, D, s) \cdot (r', D', s') = D \cdot D' - rs' - r's.$$

This is the K3 Mukai lattice. It is even, unimodular, and has signature $(4, 20)$. In the full $N = 4$ charge theory one may enlarge this to the Narain lattice $\Gamma^{6,22}$, but for the algebraic D-brane sector of $K3 \times E$, $H(S, \mathbb{Z})$ is the correct geometric starting point. Springer Link

Choose the standard basis

$$1_E \in H^0(E, \mathbb{Z}), \qquad \omega_E \in H^2(E, \mathbb{Z}), \qquad \int_E \omega_E = 1.$$

Since $H^{\text{odd}}(S) = 0$,

$$H^{\text{even}}(S \times E, \mathbb{Z}) \cong H(S, \mathbb{Z}) \otimes H^{\text{even}}(E, \mathbb{Z}).$$

Therefore every even RR/Mukai charge can be written uniquely as

$$v_X(F) = P(F) \otimes 1_E + Q(F) \otimes \omega_E,$$

with

$$P(F), Q(F) \in \Lambda_S.$$

Thus define the **dyonic Mukai charge lattice**

$$\Gamma_X^{\mathrm{phys}} := \Lambda_S \oplus \Lambda_S.$$

An object F has physical charge

$$\mathrm{ch}_{\mathrm{phys}}(F) = (Q(F), P(F)) \in \Gamma_X^{\mathrm{phys}}.$$

This is the additive lattice. For an exact triangle or Hall extension

$$F_1 \to F \to F_2,$$

one has

$$Q(F) = Q(F_1) + Q(F_2), \qquad P(F) = P(F_1) + P(F_2).$$

So the Hall algebra must be graded by

$$\Gamma_X^{\mathrm{phys}} = \Lambda_S \oplus \Lambda_S,$$

not directly by

$$\Gamma_{\mathrm{ind}} = \mathbb{Z}^3.$$

2. The Gram-index map

Now define the Gram map

$$\Pi : \Gamma_X^{\mathrm{phys}} \longrightarrow \Gamma_{\mathrm{ind}} \cong \mathbb{Z}^3$$

by

$$\Pi(Q, P) = \left(\frac{Q^2}{2}, Q \cdot P, \frac{P^2}{2} \right).$$

Because Λ_S is even,

$$\frac{Q^2}{2}, \frac{P^2}{2} \in \mathbb{Z}, \qquad Q \cdot P \in \mathbb{Z}.$$

Equivalently, define

$$T(Q,P) = \begin{pmatrix} Q^2/2 & Q\cdot P/2 \\ Q\cdot P/2 & P^2/2 \end{pmatrix}.$$

Then

$$2T(Q,P) = \begin{pmatrix} Q^2 & Q\cdot P \\ Q\cdot P & P^2 \end{pmatrix}.$$

This is exactly the matrix your paper calls the physical Gram matrix for a protected electric-magnetic pair. raeez.lorgat.automorphic-correc...

The discriminant is

$$\Delta_{\mathrm{gram}}(Q,P) = 4\begin{matrix} Q^2 & P^2 \\ 2 & 2 \end{matrix} - (Q\cdot P)^2 = Q^2P^2 - (Q\cdot P)^2.$$

So the Borcherds variables are not arbitrary formal variables. They are the Fourier characters of the rank-two Gram form:

$$q^n r^l s^m = \exp\bigl(2\pi i \,\mathrm{tr}(T(Q,P)Z)\bigr),$$

where

$$(n,l,m) = \Pi(Q,P).$$

3. The D6–D2–D0 computation

Now prove that this construction reproduces your OP/DT variables exactly.

Let $Y \subset X = S \times E$ be a one-dimensional subscheme with

$$[Y] = (\beta, d) \in H_2(S,\mathbb{Z}) \oplus H_2(E,\mathbb{Z}),$$

and

$$\chi(\mathcal{O}_Y) = k.$$

Here d is the elliptic degree and β is the K3 curve class. In your paper's primitive branch,

$$\beta = \beta_h = s_1 + hF, \qquad \beta_h^2 = 2h-2.$$

The ideal sheaf I_Y has

$$\mathrm{ch}(I_Y) = 1 - \mathrm{PD}[Y] - k[\mathrm{pt}_X].$$

Because

$$\mathrm{td}(S) = (1, 0, 1), \qquad \mathrm{td}(E) = 1,$$

the Mukai vector is

$$v_X(I_Y) = \mathrm{ch}(I_Y) \quad \mathrm{td}(X).$$

Now decompose $\mathrm{PD}[Y]$. The K3-vertical curve $\beta \times \{\mathrm{pt}\}$ contributes

$$\beta^\vee \otimes \omega_E,$$

and the elliptic curve class $d\,\{\mathrm{pt}\} \times E$ contributes

$$d[\mathrm{pt}_S] \otimes 1_E.$$

Therefore

$$v_X(I_Y) = (1, 0, 1 - d) \otimes 1_E + (0, -\beta, -k) \otimes \omega_E.$$

So set

$$P_Y = (1, 0, 1 - d), \qquad Q_Y = (0, -\beta, -k).$$

Then

$$Q_Y^2 = (-\beta)^2 = \beta^2 = 2h - 2,$$

so

$$\frac{Q_Y^2}{2} = h - 1.$$

Also

$$P_Y^2 = 0 - 2(1)(1 - d) = 2(d - 1),$$

so

$$\frac{P_Y^2}{2} = d - 1.$$

Finally,

$$Q_Y \cdot P_Y = (-\beta) \cdot 0 - 0(1 - d) - 1(-k) = k.$$

Hence

$$\Pi(Q_Y, P_Y) = (h-1,\ k,\ d-1).$$

This is exactly the OP/Borcherds variable dictionary:

$$q_{\mathrm{Bch}}^{h-1} r_{\mathrm{Bch}}^{k} s_{\mathrm{Bch}}^{d-1}.$$

So the shifts $h-1$ and $d-1$ are not ad hoc. They are the Mukai self-pairings of the two Künneth components of the D6–D2–D0 charge.

In particular,

$$\text{K3 norm} = \frac{Q_Y^2}{2} = h-1,$$

$$\text{D0/Fourier index} = Q_Y \cdot P_Y = k,$$

$$\text{elliptic degree} = \frac{P_Y^2}{2} = d-1.$$

This gives the physical derivation of your (n, l, m) variables.

4. The additive defect and its exact cocycle

Now prove why Γ_{ind} is not the Hall charge lattice.

Take two physical charges

$$c_1 = (Q_1, P_1), \qquad c_2 = (Q_2, P_2).$$

The physical sum is

$$c_1 + c_2 = (Q_1 + Q_2,\ P_1 + P_2).$$

Then

$$\Pi(c_1 + c_2) = \Pi(c_1) + \Pi(c_2) + B(c_1, c_2),$$

where the cross-term is

$$B(c_1, c_2) = (Q_1 \cdot Q_2,\ Q_1 \cdot P_2 + Q_2 \cdot P_1,\ P_1 \cdot P_2)\,.$$

Proof:

$$\frac{(Q_1+Q_2)^2}{2} = \frac{Q_1^2}{2} + \frac{Q_2^2}{2} + Q_1 \cdot Q_2,$$

$$(Q_1+Q_2)\cdot(P_1+P_2) = Q_1\cdot P_1 + Q_2\cdot P_2 + Q_1\cdot P_2 + Q_2\cdot P_1,$$

$$\frac{(P_1+P_2)^2}{2} = \frac{P_1^2}{2} + \frac{P_2^2}{2} + P_1\cdot P_2.$$

Thus

$$\Pi \text{ is quadratic, not additive.}$$

This is not a failure. It is the key structural fact.

The Hall algebra is graded by

$$\Gamma_X^{\mathrm{phys}}.$$

The Igusa/Borcherds product is graded by the invariant shadow

$$\Gamma_{\mathrm{ind}}.$$

The descent from the former to the latter must account for the cocycle B.

5. The Gram-corrected charge extension

We can package the defect as an honest additive object.

Define

$$\Gamma_X := \Gamma_X^{\mathrm{phys}} \oplus \Gamma_{\mathrm{ind}}.$$

For $c, c' \in \Gamma_X^{\mathrm{phys}}$ and $T, T' \in \Gamma_{\mathrm{ind}}$, define

$$(c,T)\star(c',T') = \big(c+c',\ T+T'+B(c,c')\big).$$

Because B is symmetric bilinear, this product is commutative and associative. Indeed,

$$B(c+c',c'') = B(c,c'') + B(c',c''),$$

so

$$((c,T)\star(c',T'))\star(c'',T'') = (c+c'+c'',\ T+T'+T''+B(c,c')+B(c,c'')+B(c',$$

and the same expression is obtained from

$$(c, T) \star ((c', T') \star (c'', T'')).$$

The identity is $(0, 0)$. The inverse of (c, T) is

$$(-c,\ -T + B(c, c)).$$

Now define

$$s : \Gamma_X^{\mathrm{phys}} \longrightarrow \Gamma_X$$

by

$$s(c) = (c, \Pi(c)).$$

Then

$$s(c) \star s(c') = (c, \Pi(c)) \star (c', \Pi(c')) = (c + c', \Pi(c) + \Pi(c') + B(c, c')) = (c + c', \Pi(c$$

Therefore

$$s : \Gamma_X^{\mathrm{phys}} \hookrightarrow \Gamma_X \text{ is an additive embedding.}$$

This is the precise mathematical repair of the non-additivity problem.

The physical charge lattice remains Γ_X^{phys}. The Gram-index lattice becomes a quadratic invariant, and the obstruction to additivity is the explicit bilinear cocycle B. The central extension Γ_X is the minimal bookkeeping object that makes the Gram data functorial under physical charge addition.

6. Primitive torsion-one descent

Now we need the invariant quotient.

Let

$$\Lambda = \Lambda_S$$

or, in the full $N = 4$ theory,

$$\Lambda = \Gamma^{6,22}.$$

Let

$$c = (Q, P) \in \Lambda \oplus \Lambda.$$

Define the rank-two charge sublattice

$$M_c := \mathbb{Z}Q + \mathbb{Z}P \subset \Lambda.$$

The Gram form on M_c is

$$G_c = \begin{pmatrix} Q^2 & Q \cdot P \\ Q \cdot P & P^2 \end{pmatrix} = 2T(\Pi(c)).$$

The pair $c = (Q, P)$ is **primitive saturated**, or torsion-one, if

$$M_c = \Lambda \cap (M_c \otimes \mathbb{Q}).$$

Equivalently, in a coordinate basis this is the condition usually written

$$\gcd(Q \wedge P) = 1.$$

The original DVV Igusa-counting formula is the torsion-one dyon formula; torsion > 1 dyons require extra discrete data beyond $Q^2, P^2, Q \cdot P$. arXiv +1

Now use the lattice theorem.

For an even unimodular indefinite lattice Λ with enough hyperbolic planes, primitive embeddings of a fixed even rank-two lattice M are unique up to $O(\Lambda)$ once the complement has sufficiently large rank relative to the discriminant length. The standard Nikulin criterion gives uniqueness when the positive and negative signatures have room and

$$\mathrm{rk}(\Lambda) - \mathrm{rk}(M) \geq 2 + \ell(A_M),$$

where $A_M = M^\vee / M$ is the discriminant group. For rank-two M, $\ell(A_M) \leq 2$, while $H(S, \mathbb{Z})$ has rank 24 and $\Gamma^{6,22}$ has rank 28, so the condition is overwhelmingly satisfied away from the usual boundary degeneracies. math.brown.edu

Thus, for primitive saturated pairs,

$$O(\Lambda) \backslash \{(Q, P) \in \Lambda^2 : \Pi(Q, P) = (n, l, m),\ \gcd(Q \wedge P) = 1\}$$

is a single orbit, subject to the standard signature and chamber conditions.

Therefore the protected index in the primitive torsion-one sector descends:

$$\Omega(Q,P) = \Omega(n,l,m), \qquad (n,l,m) = \Pi(Q,P).$$

This is the mathematical meaning of the Igusa variables:

$$\Gamma_{\mathrm{ind}} = \mathrm{Sym}^2(\mathbb{Z}^2)_{\mathrm{half}} \text{ is the primitive T-duality-invariant quotient of } \Gamma_X^{\mathrm{phys}}.$$

For torsion > 1, this is false unless one augments Γ_{ind} by the torsion invariant.

7. The Hall object should be rewritten

The compact Hall object should not be assumed to have numerical lattice Γ_{ind}. It should be:

$$H_X = \bigoplus_{c \in \Gamma_X^{\mathrm{phys}}} H_{X,c}, \qquad \Gamma_X^{\mathrm{phys}} = \Lambda_S \oplus \Lambda_S.$$

Then define the Gram pushforward only after taking the protected, duality-invariant, primitive/torsion-one sector:

$$(\Pi_* H_X)_T := \bigoplus_{\substack{[c] \in O(\Lambda_S)\backslash \Gamma_X^{\mathrm{phys}} \\ \Pi(c) = T}} H_{X,c}, \qquad T = (n,l,m).$$

In the primitive saturated sector, the orbit theorem reduces this to one orbit:

$$(\Pi_* H_X)_{(n,l,m)} \simeq H_{X,c} \quad \text{for any } c \text{ with } \Pi(c) = (n,l,m).$$

Then the desired primitive protected statement becomes

$$\mathrm{sdim}\, H^0\, \mathrm{Prim}^{\mathrm{prot}} (\Pi_* F_X)_{(n,l,m)} = f(nm, l).$$

This is the correct replacement for saying directly that the compact Hall category is Γ_{ind} -graded.

The actual Hall bracket lives first as

$$[\,,\,] : \mathrm{Prim}^{\mathrm{prot}}_{c_1} \otimes \mathrm{Prim}^{\mathrm{prot}}_{c_2} \longrightarrow \mathrm{Prim}^{\mathrm{prot}}_{c_1+c_2},$$

with

$$c_i \in \Gamma_X^{\mathrm{phys}}.$$

Only after Gram descent does one compare with the BKM bracket:

$$[\,,\,]_{\mathrm{BKM}} : \mathfrak{g}_{\alpha(n,l,m)} \otimes \mathfrak{g}_{\alpha(n',l',m')} \longrightarrow \mathfrak{g}_{\alpha(n+n',l+l',m+m')}.$$

The price of descent is exactly the cocycle $B(c_1, c_2)$. So the compact realization must prove one of the following:

$$B(c_1, c_2) = 0$$

for the primitive collision channels contributing to the BKM bracket, or else it must prove that the Hall bracket descends through the Gram-corrected extension Γ_X and then collapses to the BKM root grading.

This is now a precise mathematical problem, not a vague defect.

8. What this proves for the whitepaper

The physical charge lattice problem can now be stated as a theorem.

Theorem: Physical charge lattice and Gram descent for $K3 \times E$

Let $X = S \times E$, with S a K3 surface. The additive physical D-brane charge lattice in the even RR sector is

$$\Gamma_X^{\mathrm{phys}} = H(S, \mathbb{Z}) \oplus H(S, \mathbb{Z}).$$

The Igusa index lattice

$$\Gamma_{\mathrm{ind}} \cong \mathbb{Z}^3$$

is the image of the quadratic Gram map

$$\Pi(Q, P) = \left(\frac{Q^2}{2}, Q \cdot P, \frac{P^2}{2} \right).$$

For a rank-one D6–D2–D0 ideal-sheaf charge with curve class (β, d), Euler characteristic k, and $\beta^2 = 2h - 2$,

$$v_X(I_Y) = (1, 0, 1 - d) \otimes 1_E + (0, -\beta, -k) \otimes \omega_E.$$

Choosing the Igusa electric-magnetic pair

$$Q_Y = (0, -\beta, -k), \qquad P_Y = (1, 0, 1-d),$$

one obtains

$$\Pi(Q_Y, P_Y) = (h-1, k, d-1).$$

Thus the OP/Borcherds monomial is exactly

$$q^{h-1} r^k s^{d-1} = \exp\big(2\pi i \operatorname{tr}(T(Q_Y, P_Y) Z)\big).$$

The map Π is not additive. Its failure is the explicit bilinear cocycle

$$B((Q_1, P_1), (Q_2, P_2)) = (Q_1 \cdot Q_2,\ Q_1 \cdot P_2 + Q_2 \cdot P_1,\ P_1 \cdot P_2).$$

The Gram-corrected extension

$$\Gamma_X = \Gamma_X^{\mathrm{phys}} \oplus \Gamma_{\mathrm{ind}}$$

with multiplication

$$(c, T) \star (c', T') = (c + c', T + T' + B(c, c'))$$

makes the section

$$c \mapsto (c, \Pi(c))$$

additive. Hence the Hall category is physically graded by Γ_X^{phys}, while the Igusa/Borcherds denominator is the primitive torsion-one $O(\Lambda_S)$-invariant Gram descent of that grading.

9. The exact paper-level replacement

Replace the sentence

$$\text{“Suppose an extension-closed compact Hall category with numerical Gram-index la}$$

by:

$$\text{“Suppose an extension-closed compact Hall category graded by } \Gamma_X^{\mathrm{phys}} = H(S, \mathbb{Z}) \oplus$$

Then the recognition target becomes

$$H^0 \Pi_* \operatorname{Prim}^{\mathrm{prot}}(F_X) \cong \mathfrak{g}_{\Delta_5}.$$

Not

$$H^0\,\mathrm{Prim}^{\mathrm{prot}}(F_X) \cong \mathfrak{g}_{\Delta_5}$$

before descent.

That is the crucial correction.

10. The physics interpretation

The two Künneth Mukai components

$$Q = (0, -\beta, -k), \qquad P = (1, 0, 1-d)$$

are the two physical charge vectors whose Gram invariants are counted by the Igusa variables.

The K3 curve norm gives

$$Q^2/2 = h - 1.$$

The D0/Fourier charge gives

$$Q \cdot P = k.$$

The elliptic degree, with the Todd/vacuum shift, gives

$$P^2/2 = d - 1.$$

So the Igusa triple is physically

$$(n, l, m) = \left(\frac{Q^2}{2}, Q \cdot P, \frac{P^2}{2}\right) = (h-1, k, d-1).$$

This is exactly the same triple that appears in the Borcherds product

$$\prod_{(n,l,m)>0} (1 - q^n r^l s^m)^{f(nm,l)}.$$

The product does not live directly on the raw physical lattice. It lives on the primitive Gram quotient of the physical lattice.

That is the construction.

Sources

Sources

I want it to be a dirac-inspired paper (as if it was written by dirac) on the physics here, that makes natural the factorisation algebra from physical first-principles etc. locality (or lack thereof) is treated physically. unitarity or lack thereof...

I'll recast it in a Dirac style: begin with observables, charges, locality, and the protected supercharge, then derive factorization, Hall structure, and the Igusa square root.

The crucial shift is that charged BPS states are not local fields; locality governs protected observables and collision laws before charge completion and trace reveal the automorphic product.

Thought for 6m 51s

Yes. The paper should be **Dirac-inspired** in exactly this sense:

> Do not begin with a scalar partition function and ask for a square root.
> Begin with a first-order physical object whose scalar shadow is a square.

That is the Dirac move. Dirac did not take an analytic square root of the Klein–Gordon operator. He introduced new algebraic degrees of freedom — spinors and Clifford multiplication — such that the square became scalar. Your paper should do the same. The Igusa square root should not be presented as a branch of Δ_5^2. It should be presented as the **first-order protected chiral/Pfaffian object** whose scalar reduced DT/OP theory is the square. Your current manuscript already gestures at this through protected locality, the E_3-source, the orientation character, and the statement that the scalar square forgets the automorphic reflection sign. The rewrite should make that the central physics rather than a caveat. raeez.lorgat.automorphic-correc...

Here is the spine I would build.

The Dirac principle for the Igusa square root

The scalar theory gives

$$Z^X_{\text{scalar}} = -4096\,\Delta_5(Z)^{-2}.$$

This is the analogue of the second-order equation. It is invariant under the sign character because the sign has been squared away.

The first-order theory is not

$$Z^X_{\text{scalar}}$$

as an analytic function. It is a protected chiral object

$$\mathcal{A}_X$$

with a determinant/Pfaffian

$$\mathrm{Pf}_{\text{prot}}(\mathcal{A}_X) = \Delta_5(Z),$$

and whose scalar norm or doubled trace gives

$$Z^X_{\text{scalar}} = C_{\text{scalar}}\ \mathrm{Pf}_{\text{prot}}(\mathcal{A}_X)^{-2}.$$

For the OP/DT normalization,

$$C_{\text{scalar}} = -4096.$$

Thus the actual problem is not:

$$\text{choose a square root of } \Delta_5^2.$$

It is:

$$\text{construct the first-order protected algebra whose determinant is } \Delta_5.$$

This is the Dirac formulation.

The new degrees of freedom replacing “spinors” are:

$$\text{orientation line} + \text{primitive Hall bracket} + \text{chiral/factorization locality} + \text{Borcherc}$$

The scalar square forgets them. The square root remembers them.

1. States are not first. Observables are first.

A charged BPS state is not a local field. It is a sector. Its charge is measured at infinity. Therefore the microscopic Hilbert space is not the natural first object.

The first object is the algebra of protected operations.

A state is a representation of that algebra. A partition function is a trace of that representation. A trace cannot reconstruct the algebra. So the paper should not begin with the scalar trace and then apologize that it cannot reconstruct locality. It should begin with locality and then show what trace it produces.

The physical hierarchy should be:

$$\text{local protected operations} \longrightarrow \text{factorization/Hall algebra} \longrightarrow \text{primitive Lie algebra}$$

This is already latent in your manuscript, but it should become the philosophical and technical backbone.

2. The physical charge lattice

Let

$$X = S \times E,$$

where S is a K3 surface.

The even Mukai lattice of S is

$$\Lambda_S = H(S, \mathbb{Z}) = H^0(S, \mathbb{Z}) \oplus H^2(S, \mathbb{Z}) \oplus H^4(S, \mathbb{Z}),$$

with pairing

$$(r, D, s) \cdot (r', D', s') = D \cdot D' - rs' - r's.$$

Since

$$H^{\text{even}}(X, \mathbb{Z}) \cong H(S, \mathbb{Z}) \otimes H^{\text{even}}(E, \mathbb{Z}),$$

every even charge decomposes as

$$v_X(F) = P(F) \otimes 1_E + Q(F) \otimes \omega_E.$$

Thus the additive physical charge lattice is

$$\Gamma_X^{\mathrm{phys}} = \Lambda_S \oplus \Lambda_S,$$

with charge

$$c(F) = (Q(F), P(F)).$$

This is the lattice on which Hall multiplication is genuinely additive.

The Igusa variables are not this lattice. They are the Gram shadow:

$$\Pi(Q,P) = \left(\frac{Q^2}{2}, Q\cdot P, \frac{P^2}{2}\right) = (n,l,m).$$

So the Fourier monomial is

$$q^n r^l s^m = \exp\left(2\pi i \,\mathrm{tr}(T(Q,P)Z)\right),$$

where

$$T(Q,P) = \begin{pmatrix} Q^2/2 & Q\cdot P/2 \\ Q\cdot P/2 & P^2/2 \end{pmatrix}.$$

Thus the genus-two variable Z is a chemical potential for the rank-two charge plane spanned by Q and P.

For the D6–D2–D0 ideal-sheaf sector,

$$v_X(I_Y) = (1,0,1-d)\otimes 1_E + (0,-\beta,-k)\otimes \omega_E.$$

So

$$P_Y = (1,0,1-d), \qquad Q_Y = (0,-\beta,-k).$$

If

$$\beta^2 = 2h-2,$$

then

$$\frac{Q_Y^2}{2} = h-1,$$

$$Q_Y\cdot P_Y = k,$$

$$\frac{P_Y^2}{2} = d - 1.$$

Therefore

$$\Pi(Q_Y, P_Y) = (h-1, k, d-1).$$

This is the physical derivation of the Borcherds/OP variables.

The paper should state this early. It makes (n, l, m) natural without pretending that $\mathbb{Z}^3$ is the microscopic charge lattice.

3. The local object on X

The local object is not the scalar DT series. The local object is the protected open-string observable algebra on the Calabi–Yau threefold X.

In the B-model language, the local field theory on a Calabi–Yau threefold is holomorphic Chern–Simons/BV theory. Locally, for a brane background F, the fields are controlled by the dg Lie algebra

$$\mathfrak{g}_F(U) = \mathrm{RHom}_U(F, F)[1]$$

or, perturbatively,

$$\Omega_c^{0,*}(U, \mathrm{End}\, F)[1].$$

The BV pairing comes from Serre duality and the holomorphic volume form on X. The classical interaction is cubic:

$$S(A) = \int_X \Omega_X \wedge \mathrm{Tr}\left(\frac{1}{2} A\bar{\partial} A + \frac{1}{3} A^3\right).$$

The compact nonperturbative version replaces one fixed brane by the derived stack of compactly supported branes.

For an open $U \subset X$, let

$$\mathfrak{M}_c(U)$$

be the derived stack of compactly supported coherent B-branes in U with physical charge

$$c \in \Gamma_X^{\mathrm{phys}}.$$

The Calabi–Yau threefold structure gives this stack a (-1)-shifted symplectic structure. After choosing an orientation line $\mathfrak{o}_c$, define the protected local object

$$\mathcal{A}_X(U) = \bigoplus_{c \in \Gamma_X^{\mathrm{phys}}} H_*^{BM}\left(\mathfrak{M}_c(U), \Phi_c \otimes \mathfrak{o}_c\right).$$

Here Φ_c denotes the vanishing-cycle/perverse-sheaf coefficient system coming from the local critical model, and the completion is the physical charge/HN completion.

This is the object the paper should call the compact protected observable algebra.

It is not yet a Hilbert space. It is a cohomological algebra of protected operations.

4. Why this is naturally an E_3-factorization algebra

The reason is purely local.

Let $U_1, \dots, U_k \subset V \subset X$ be pairwise disjoint holomorphic opens. If

$$F_i \in \mathfrak{M}_{c_i}(U_i),$$

then

$$F_1 \oplus \cdots \oplus F_k \in \mathfrak{M}_{c_1+\cdots+c_k}(V).$$

Because the supports are disjoint,

$$\mathrm{Ext}^\bullet(F_i, F_j) = 0 \qquad (i \neq j).$$

Therefore the deformation-obstruction theory splits:

$$\mathrm{RHom}\left(\bigoplus_i F_i, \bigoplus_i F_i\right) \simeq \bigoplus_i \mathrm{RHom}(F_i, F_i).$$

The critical functions add, the vanishing cycles factor by Thom–Sebastiani, and the orientation lines multiply:

$$\Phi_{\oplus_i F_i} \simeq \boxtimes_i \Phi_{F_i},$$

$$\mathfrak{o}_{\oplus_i F_i} \simeq \boxtimes_i \mathfrak{o}_{F_i}.$$

Hence one obtains factorization maps

$$\mathcal{A}_X(U_1) \otimes \cdots \otimes \mathcal{A}_X(U_k) \longrightarrow \mathcal{A}_X(V).$$

When the opens collide, the cross-Ext groups no longer vanish. Their nonzero extensions are exactly the Hall correspondences:

$$\mathfrak{M}_{c_1} \times \mathfrak{M}_{c_2} \longleftarrow \mathfrak{E}_{c_1,c_2} \longrightarrow \mathfrak{M}_{c_1+c_2}.$$

Thus disjointness gives ordinary factorization; collision gives the Hall product.

This is the physical origin of the factorization algebra.

The reason the structure is called holomorphic E_3, rather than real E_6, is that the protected supercharge has killed the anti-holomorphic translations. The remaining local singularities depend on three complex holomorphic directions. A small holomorphic polydisc in X behaves like the local model for a holomorphic E_3-algebra. With a choice of formality/framing, one obtains an ordinary E_3-shadow.

So the paper should say:

$$\mathcal{A}_X \in \mathrm{Fact}^{\mathrm{hol}}_{E_3}(X)$$

means:

> the protected local observables of the holomorphic Calabi–Yau threefold theory form a holomorphic factorization algebra whose pointwise local model is an E_3-algebra after the chosen formality/framing.

This makes the E_3 natural, not decorative.

5. $K3 \to E$ specialization

Now project

$$p_E : X = S \times E \to E.$$

For an open $V \subset E$, define the projection-finite sector

$$\mathfrak{M}^{\mathrm{pf}}_c(V) = \{F \in \mathfrak{M}_c(S \times V) \ : \ p_E(\mathrm{Supp}\, F) \subset V \text{ is finite}\}\,.$$

Then

$$\mathcal{F}_X^{\mathrm{pf}}(V) = \bigoplus_c H_*^{BM}\left(\mathfrak{M}_c^{\mathrm{pf}}(V), \Phi_c \otimes \mathfrak{o}_c\right)$$

is a genuine chiral/factorization object on $\mathrm{Ran}(E)$.

For disjoint $V_i \subset V$, projection-finite supports remain disjoint, so one has

$$\mathcal{F}_X^{\mathrm{pf}}(V_1) \otimes \cdots \otimes \mathcal{F}_X^{\mathrm{pf}}(V_k) \longrightarrow \mathcal{F}_X^{\mathrm{pf}}(V).$$

Collisions in E remember extensions inside $S \times E$. Therefore the chiral algebra on E is not imposed by hand. It is the $K3$-integrated shadow of the local E_3-object on X.

Symbolically,

$$\mathcal{F}_X^{\mathrm{pf}} \simeq \mathrm{Sp}_{S,E}^{\mathrm{ch}}(\mathcal{A}_X)_{\mathrm{pf}}.$$

This is the correct physical meaning of the $K3 \to E$ specialization.

6. Locality fails exactly where it should fail

Now the important point: positive elliptic degree is not local on E.

Let $C \subset S \times E$ be a one-dimensional subscheme with positive elliptic degree:

$$(p_E)_*[C] = d[E], \qquad d > 0.$$

Then $p_E(C)$ is a one-dimensional closed subset of the irreducible curve E. Hence

$$p_E(C) = E.$$

Therefore C cannot be assigned to finitely many points of E.

This is not a mathematical inconvenience. It is the physics.

A brane wrapping E is not a local insertion on E. It is a global sector, just as a Wilson loop is not a pointlike local observable in the direction it wraps. It can couple to local insertions, but it is not itself localized at their points.

Therefore the correct object is not simply

$$\mathrm{Fact}(\mathrm{Ran}(E)).$$

It is a hybrid object:

$$\mathcal{F}_X^{\mathrm{hyb}} = \mathcal{F}_X^{\mathrm{pf}} \curvearrowright \mathcal{W}_X.$$

Here

$$\mathcal{W}_X = \bigoplus_{m>0} \mathcal{W}_{X,m}$$

is the wrapped elliptic-degree sector, with

$$m = \frac{P^2}{2} = d - 1.$$

The projection-finite factorization algebra acts on the wrapped sector by mixed Hall correspondences:

$$\mathcal{F}_X^{\mathrm{pf}}(V_1) \otimes \cdots \otimes \mathcal{F}_X^{\mathrm{pf}}(V_k) \otimes \mathcal{W}_{X,m} \longrightarrow \mathcal{W}_{X,m'}.$$

Wrapped-wrapped extensions give

$$\mathcal{W}_{X,m} \otimes \mathcal{W}_{X,m'} \longrightarrow \mathcal{W}_{X,m+m'+\mathrm{interaction}}.$$

The interaction term is measured on the additive physical charge lattice before Gram descent.

This is the physical replacement for the naive Ran(E) locality statement.

The s-direction in the Igusa product is therefore not a local point-support direction on E. It is the wrapped/global direction. The factorization algebra survives, but it becomes **stratified**:

$$\text{local vertical sector} + \text{global wrapped sector} + \text{mixed Hall action}.$$

That is the correct locality physics.

7. The Hall bracket and primitive algebra

The extension correspondence is

$$\mathfrak{M}_c \times \mathfrak{M}_{c'} \longleftarrow \mathfrak{E}_{c,c'} \longrightarrow \mathfrak{M}_{c+c'}.$$

After orientation and protected integration, it gives a Hall product

$$* : H_c \otimes H_{c'} \longrightarrow H_{c+c'}.$$

The primitive part of the completed Hall coalgebra gives a Lie superalgebra:

$$[\,,\,] : \mathrm{Prim}_c^{\mathrm{prot}} \otimes \mathrm{Prim}_{c'}^{\mathrm{prot}} \longrightarrow \mathrm{Prim}_{c+c'}^{\mathrm{prot}}.$$

This bracket is genuinely graded by

$$\Gamma_X^{\mathrm{phys}} = \Lambda_S \oplus \Lambda_S.$$

The Igusa/Borcherds algebra is obtained only after Gram descent:

$$\Pi(Q,P) = \left(\frac{Q^2}{2}, Q\cdot P, \frac{P^2}{2}\right).$$

Thus the comparison should be

$$H^0 \Pi_* \mathrm{Prim}^{\mathrm{prot}}(\mathcal{F}_X^{\mathrm{hyb}}) \cong \mathfrak{g}_{\Delta_5}.$$

Not

$$H^0 \mathrm{Prim}^{\mathrm{prot}}(\mathcal{F}_X) \cong \mathfrak{g}_{\Delta_5}$$

before descent.

The descent is physical: the full charge lattice is microscopic, but the protected automorphic trace sees only the duality-invariant Gram plane.

8. The Dirac square root and the orientation character

The scalar square is insensitive to the sign character:

$$\Delta_5^2$$

is scalar.

But the square root

$$\Delta_5$$

transforms with the Maass character

$$\nu_{\Delta_5}.$$

Physically this character must come from the square root of a determinant line. That is the exact analogue of a Pfaffian.

Let $\mathscr{L}_c$ be the determinant line of the deformation complex at charge c:

$$\mathscr{L}_c = \det \mathrm{RHom}(F, F).$$

An orientation is a square root

$$\mathfrak{o}_c^{\otimes 2} \simeq \mathscr{L}_c.$$

A wall reflection preserves $\mathscr{L}_c$, but it may reverse the square root $\mathfrak{o}_c$. Therefore the scalar determinant is invariant, while the Pfaffian changes sign.

For the three type-II simple reflections

$$s_{\delta_1}, \quad s_{\delta_2}, \quad s_{\delta_3},$$

the expected local wall model has one real normal mode. Hence

$$\epsilon_o(s_{\delta_i}) = -1, \qquad i = 1, 2, 3.$$

The scalar square sees

$$(-1)^2 = 1.$$

The square root sees

$$-1.$$

Therefore the orientation-monodromy character should satisfy

$$\epsilon_o|_{W^{(2)}(\Lambda_{II}^{2,1})} = \nu_{\Delta_5}|_{W^{(2)}(\Lambda_{II}^{2,1})}.$$

This is not a normalization issue. It is the physical sign of the first-order theory.

In Dirac language:

$$\Delta_5^{-2}$$

is the scalar second-order object.

$$\Delta_5^{-1}$$

is the chiral first-order object.

The passage from the second to the first requires an orientation/Pfaffian datum, just as the passage from Klein–Gordon to Dirac requires spinor/Clifford data.

9. Unitarity is used before the index, not after it

The paper should be very explicit here.

The untwisted theory is unitary. This matters because it gives an adjoint supercharge:

$$Q^\dagger,$$

and an identity of the form

$$\{Q, Q^\dagger\} = H_{\text{right}}.$$

Then all states with positive right-moving energy pair and cancel in the protected supertrace. This is why the K3 elliptic genus localizes.

So unitarity supplies the cancellation theorem.

But after taking Q-cohomology, one no longer has a positive Hilbert space. One has a cohomological, charge-graded, virtual object.

Thus

$$f(n, l)$$

is a superdimension, not a dimension.

A negative coefficient means an odd excess:

$$f(n, l) < 0 \quad \Longleftrightarrow \quad \dim H_{\bar{1}} - \dim H_{\bar{0}} > 0$$

in whatever protected representative realizes it.

The Borcherds denominator algebra is also not a unitary Hilbert-space representation. Its Cartan lattice is Lorentzian. Its multiplicities are signed supermultiplicities:

$$\operatorname{smult}(\alpha) = \dim \mathfrak{g}_{\alpha,\bar{0}} - \dim \mathfrak{g}_{\alpha,\bar{1}}.$$

This is exactly what one expects from a protected cohomological theory.

So the correct statement is:

unitarity is an input to localization; the square root is a cohomological protected

That is not a weakness. It is the nature of an index.

10. The determinant as the protected first-order amplitude

Assume the primitive protected algebra exists and Gram descends to

$$\Pi_* \mathrm{Prim}^{\mathrm{prot}} = \bigoplus_{(n,l,m)} P_{n,l,m}.$$

The Dirac/IQFT requirement is

$$\mathrm{sdim}\, P_{n,l,m} = f(nm, l).$$

Then the first-order protected determinant is

$$D_X(Z) = 64 q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)>0} (1 - q^n r^l s^m)^{\mathrm{sdim}\, P_{n,l,m}}.$$

Hence

$$D_X(Z) = 64 q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)>0} (1 - q^n r^l s^m)^{f(nm,l)} = \Delta_5(Z).$$

This is the protected Pfaffian/determinant of the first-order object.

The scalar OP/DT theory gives

$$Z^X_{\mathrm{OP/DT}} = -4096\, \Delta_5^{-2} = -4096\, D_X^{-2}.$$

So the OP/DT scalar theory is the squared, orientation-forgetting theory.

This is the clean physical meaning of the Igusa square root.

11. The denominator algebra as the first-order equation

In the Dirac equation, the first-order operator is not arbitrary. Its square gives the scalar operator because the gamma matrices obey a Clifford relation.

Here the analogue is the Borcherds–Kac–Moody denominator algebra.

The primitive algebra should be generated by real and imaginary simple protected operations, with the real simple roots

$$\delta_1 = 2f_2 - f_3, \qquad \delta_2 = 2f_{-2} - f_3, \qquad \delta_3 = f_3,$$

and Cartan matrix

$$A = \begin{matrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{matrix}.$$

The first-order algebraic equation is the denominator identity

$$\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z).$$

The scalar square follows only after taking the determinant square.

So the physical recognition theorem should be written as:

$$\text{The compact protected primitive algebra is the Dirac operator.}$$

More precisely,

$$H^0\Pi_*\mathrm{Prim}^{\mathrm{prot}}(\mathcal{F}_X^{\mathrm{hyb}}) \cong \mathfrak{g}_{\Delta_5}.$$

The denominator identity is then the first-order spectral equation of the protected algebra.

12. Proposed replacement thesis paragraph

I would put something like this near the beginning:

> The square root in this paper is meant in Dirac's sense. A scalar square is not square-rooted by choosing a branch. It is square-rooted by finding a first-order algebra whose determinant, after forgetting orientation, gives the scalar object. For $X = K3 \times E$, the scalar reduced DT/OP branch gives $-4096\Delta_5^{-2}$. The first-order object sought here is a protected holomorphic E_3-factorization algebra on X, with a hybrid chiral shadow on E, whose primitive Hall bracket Gram-descends to the Gritsenko–Nikulin Borcherds–Kac–Moody algebra $\mathfrak{g}_{\Delta_5}$. The scalar square forgets the Pfaffian orientation character; the square root remembers it as the Maass character

of Δ_5. Thus the problem is not analytic extraction of Δ_5 from Δ_5^2, but construction of the protected first-order algebra whose determinant is Δ_5.

That gives the whole paper a physical mission.

13. Proposed definition: Dirac–Igusa datum

Replace the weaker "compact Igusa realization datum" with a sharper physical package.

A **Dirac–Igusa datum** for $X = S \times E$ consists of:

$$\mathfrak{D}_X = \left(\Gamma_X^{\mathrm{phys}}, \mathcal{A}_X, \mathcal{F}_X^{\mathrm{hyb}}, \mathfrak{o}, \Pi, \epsilon_o, \Theta_{\mathrm{den}}\right),$$

where:

$$\Gamma_X^{\mathrm{phys}} = H(S,\mathbb{Z}) \oplus H(S,\mathbb{Z})$$

is the additive physical charge lattice;

$$\mathcal{A}_X \in \mathrm{Fact}_{E_3}^{\mathrm{hol}}(X)$$

is the charge-completed protected local observable algebra;

$$\mathcal{F}_X^{\mathrm{hyb}} = \mathcal{F}_X^{\mathrm{pf}} \curvearrowright \mathcal{W}_X$$

is the hybrid $K3 \to E$ chiral shadow with projection-finite local sector and wrapped elliptic-degree sector;

$$\mathfrak{o}_c^{\otimes 2} \simeq \det \mathrm{RHom}(F,F)$$

is the orientation/Pfaffian datum;

$$\Pi(Q,P) = \left(\frac{Q^2}{2}, Q\cdot P, \frac{P^2}{2}\right)$$

is the Gram descent;

$$\epsilon_o : W^{(2)}(\Lambda_{II}^{2,1}) \to \{\pm 1\}$$

is the induced orientation monodromy, required to satisfy

$$\epsilon_o = \nu_{\Delta_5}$$

on the type-II Weyl group;

and

$$\Theta_{\mathrm{den}} : H^0 \Pi_* \mathrm{Prim}^{\mathrm{prot}}(\mathcal{F}_X^{\mathrm{hyb}}) \quad \sim \quad \mathfrak{g}_{\Delta_5}$$

is the primitive denominator comparison.

Then the protected determinant is

$$D_X(Z) = \mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X) = \Delta_5(Z),$$

and the scalar reduced branch is

$$Z^X_{\mathrm{OP/DT}} = -4096\, D_X(Z)^{-2}.$$

This is the Dirac theorem the paper wants.

14. Locality summary in one sentence

The paper should say:

$$\text{BPS states are not local, but protected operations on BPS sectors factorize.}$$

Then refine:

$$\text{Projection-finite sectors are local on } \mathrm{Ran}(E); \quad \text{positive elliptic-degree sectors are}$$

This makes the locality obstruction physical rather than embarrassing.

15. Unitarity summary in one sentence

The paper should say:

$$\text{unitarity proves the index localization; the index itself is a cohomological amplitud}$$

Then refine:

$$\Delta_5^{-1} \text{ is orientation-sensitive and first-order;} \qquad \Delta_5^{-2} \text{ is scalar and orientation-forge}$$

That is the physics of the square root.

16. The new title-level interpretation

The title “The Igusa Square Root” then means:

$$\text{The Igusa cusp form } \Delta_5 \text{ is the Dirac/Pfaffian half of the scalar } K3 \times E \text{ index.}$$

Not half in the sense of half a Hilbert space.

Not half in the sense of an analytic branch.

Half in the sense of a first-order protected algebra whose determinant squares to the scalar branch.

That is the paper.

Sources

Sources

Genuinely close — one provable, one a real open theorem:

1. Provable now (lemma-strength, not literature-advancing but new in the manuscript): $B(c,c') =

Show more

I’ll write this as manuscript-ready theorem/proof blocks: first the unconditional cocycle extension, then falsifiable conditional descent, then the strongest Pfaffian/orientation statement possible.

One refinement matters: the B-extension is cohomologically split, so its force is not a new charge law but canonical bookkeeping. For descent, B-isotropy becomes the exact obstruction criterion.

Thought for 8m 59s

I can prove more than item 1. The cocycle is not only associative bookkeeping; it gives the **right corrected descent map**. The important twist is this:

$$\text{Raw } \Pi\text{-descent by } B\text{-isotropy is too strong for the full BKM bracket.}$$

The full real-root Serre strings of $\mathfrak{g}_{\Delta_5}$ force repeated brackets such as

$$[e_i, [e_i, e_j]] \neq 0,$$

and those cannot remain raw-Π-additive, because $B(c_i, c_i) = 2\Pi(c_i) \neq 0$. So the candidate theorem should be sharpened:

$$\text{The compact primitive algebra should be graded by a } B\text{-twisted/normal-ordered c}$$

not by Γ_{phys} alone and not by raw Γ_{ind} alone. This is exactly the Dirac move: introduce the extra first-order datum needed to make the square-root algebra linear.

Your manuscript already has the necessary ingredients: Γ_{ind}, the Gram convention, the BKM low-degree real-root data, the open compact Hall/factorization datum, and the orientation-monodromy target $\epsilon_o = \nu_{\Delta_5}$. raeez.lorgat.automorphic-correc...

1. Physical charge lattice and Gram map

Let Λ be an even integral lattice. For the algebraic $K3 \times E$ D-brane sector take

$$\Lambda = H(S, \mathbb{Z}),$$

with Mukai pairing

$$(r, D, s) \cdot (r', D', s') = D \cdot D' - rs' - r's.$$

Define the additive physical charge lattice

$$\Gamma_X^{\text{phys}} := \Lambda \oplus \Lambda.$$

A physical charge is written

$$c = (Q, P).$$

Define the Gram-index map

$$\Pi : \Gamma_X^{\mathrm{phys}} \longrightarrow \Gamma_{\mathrm{ind}} \cong \mathbb{Z}^3, \qquad \Pi(Q,P) = \left(\frac{Q^2}{2}, Q \cdot P, \frac{P^2}{2} \right).$$

Because Λ is even, $Q^2/2$ and $P^2/2$ are integral.

2. Explicit primitive lift of every Gram triple

This is a useful first-principles lemma.

Lemma 2.1 — Surjectivity of the Gram map

Suppose Λ contains two orthogonal hyperbolic planes

$$U_1 \oplus U_2 = \mathbb{Z}e_1 \oplus \mathbb{Z}f_1 \oplus \mathbb{Z}e_2 \oplus \mathbb{Z}f_2$$

with

$$e_i^2 = f_i^2 = 0, \qquad e_i \cdot f_i = 1, \qquad U_1 \perp U_2.$$

Then for every triple

$$(n,l,m) \in \mathbb{Z}^3$$

there exists a primitive saturated physical pair $(Q,P) \in \Lambda^2$ with

$$\Pi(Q,P) = (n,l,m).$$

Explicitly, take

$$Q = e_1 + nf_1,$$

$$P = lf_1 + e_2 + mf_2.$$

Then

$$Q^2 = 2n, \qquad P^2 = 2m, \qquad Q \cdot P = l.$$

Therefore

$$\Pi(Q,P) = (n,l,m)\,.$$

The sublattice $\mathbb{Z}Q + \mathbb{Z}P \subset \Lambda$ is primitive because the coefficient matrix of Q, P in the basis e_1, f_1, e_2, f_2 contains the 2×2 minor

$$\det\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix} = 1.$$

So every formal Igusa degree has an honest primitive physical charge-plane lift.

This proves that the obstruction is **not** pointwise existence of charges. The obstruction is compatibility with Hall addition.

3. The cocycle theorem

For

$$c = (Q, P), \qquad c' = (Q', P'),$$

define

$$B(c, c') = (Q \cdot Q',\ Q \cdot P' + Q' \cdot P,\ P \cdot P') \in \Gamma_{\mathrm{ind}}.$$

This is the polarization of the quadratic Gram map.

Theorem 3.1 — B is a symmetric bilinear normalized 2-cocycle

For all $c, c', c'' \in \Gamma_X^{\mathrm{phys}}$,

$$B(c, c') = B(c', c),$$

$$B(c + c', c'') = B(c, c'') + B(c', c''),$$

$$B(c, c' + c'') = B(c, c') + B(c, c''),$$

and

$$B(c, 0) = B(0, c) = 0.$$

Moreover,

$$\Pi(c + c') = \Pi(c) + \Pi(c') + B(c, c').$$

Consequently B satisfies the additive 2-cocycle equation

$$B(c, c') + B(c + c', c'') = B(c', c'') + B(c, c' + c'').$$

Proof

Write $c = (Q, P)$, $c' = (Q', P')$, $c'' = (Q'', P'')$. Symmetry is immediate from symmetry of the lattice pairing.

Bilinearity follows because each component of B is bilinear in (Q, P) and (Q', P'). For example,

$$(Q + Q') \cdot Q'' = Q \cdot Q'' + Q' \cdot Q'',$$

and similarly for the other components.

The Gram-defect identity follows from expanding:

$$\frac{(Q+Q')^2}{2} = \frac{Q^2}{2} + \frac{Q'^2}{2} + Q \cdot Q',$$

$$(Q + Q') \cdot (P + P') = Q \cdot P + Q' \cdot P' + Q \cdot P' + Q' \cdot P,$$

$$\frac{(P+P')^2}{2} = \frac{P^2}{2} + \frac{P'^2}{2} + P \cdot P'.$$

Thus

$$\Pi(c + c') = \Pi(c) + \Pi(c') + B(c, c').$$

Finally,

$$B(c, c') + B(c + c', c'') = B(c, c') + B(c, c'') + B(c', c''),$$

while

$$B(c', c'') + B(c, c' + c'') = B(c', c'') + B(c, c') + B(c, c'').$$

The two sides agree. ∎

4. The B-twisted charge extension

Define

$$\Gamma_X := \Gamma_X^{\text{phys}} \oplus_B \Gamma_{\text{ind}}$$

as the set

$$\Gamma_X^{\text{phys}} \oplus \Gamma_{\text{ind}}$$

with multiplication

$$(c,\kappa)\star(c',\kappa') = (c+c',\ \kappa+\kappa'+B(c,c'))\,.$$

Here κ should be interpreted as the **Gram defect** or **normal-ordering counterterm**, not as the final Igusa degree.

Theorem 4.1 — Γ_X is an abelian central extension

The operation $\star$ makes Γ_X into an abelian group fitting into an exact sequence

$$0 \longrightarrow \Gamma_{\mathrm{ind}} \longrightarrow \Gamma_X \longrightarrow \Gamma_X^{\mathrm{phys}} \longrightarrow 0,$$

where

$$\Gamma_{\mathrm{ind}} \hookrightarrow \Gamma_X, \qquad \kappa \mapsto (0,\kappa),$$

and

$$\Gamma_X \to \Gamma_X^{\mathrm{phys}}, \qquad (c,\kappa)\mapsto c.$$

The identity is

$$(0,0),$$

and the inverse of (c,κ) is

$$(c,\kappa)^{-1} = (-c,\ -\kappa+B(c,c))\,.$$

Proof

Associativity follows from the cocycle identity:

$$((c,\kappa)\star(c',\kappa'))\star(c'',\kappa'') = (c+c'+c'',\ \kappa+\kappa'+\kappa''+B(c,c')+B(c+c',c'')),$$

whereas

$$(c,\kappa)\star((c',\kappa')\star(c'',\kappa'')) = (c+c'+c'',\ \kappa+\kappa'+\kappa''+B(c',c'')+B(c,c'+c'')).$$

The two second components are equal by Theorem 3.1.

Commutativity follows from symmetry of B. The identity and inverse formulas are direct. Exactness is immediate. ∎

5. The key construction: the normal-ordered Gram map

The raw Gram map

$$\Pi : \Gamma_X^{\mathrm{phys}} \to \Gamma_{\mathrm{ind}}$$

is not additive. But the B-twisted extension carries a canonical additive replacement.

Define

$$\Pi : \Gamma_X \longrightarrow \Gamma_{\mathrm{ind}}$$

by

$$\Pi(c, \kappa) = \Pi(c) - \kappa.$$

Theorem 5.1 — Π is a group homomorphism

For all $(c, \kappa), (c', \kappa') \in \Gamma_X$,

$$\Pi\big((c, \kappa) \star (c', \kappa')\big) = \Pi(c, \kappa) + \Pi(c', \kappa').$$

Proof

Using the definition of $\star$,

$$(c, \kappa) \star (c', \kappa') = (c + c', \kappa + \kappa' + B(c, c')).$$

Therefore

$$\begin{aligned}
\Pi\big((c, \kappa) \star (c', \kappa')\big) &= \Pi(c + c') - \kappa - \kappa' - B(c, c') \\
&= \Pi(c) + \Pi(c') + B(c, c') - \kappa - \kappa' - B(c, c') \\
&= (\Pi(c) - \kappa) + (\Pi(c') - \kappa') \\
&= \Pi(c, \kappa) + \Pi(c', \kappa').
\end{aligned}$$

∎

This is the main repair.

The raw physical Gram is

$$\Pi(c).$$

The additive Borcherds/Iausa degree is

$$\Pi(c,\kappa) = \Pi(c) - \kappa.$$

The defect is

$$\kappa = \Pi(c) - \Pi(c,\kappa).$$

So a bound state can have physical charge c, actual Gram $\Pi(c)$, and normal-ordered Borcherds degree $\Pi(c,\kappa)$. The difference is precisely the accumulated pairwise interaction term.

6. Flag formula: the defect is the sum of pairwise cross-Grams

Suppose $c_1, \dots, c_k \in \Gamma_X^{\mathrm{phys}}$ are elementary one-particle charges, placed initially in defect-zero degrees

$$(c_i, 0) \in \Gamma_X.$$

Then

$$(c_1,0) \star \cdots \star (c_k,0) = \left(\sum_{i=1}^{k} c_i, \sum_{1 \le i < j \le k} B(c_i,c_j) \right).$$

Therefore

$$\Pi\big((c_1,0) \star \cdots \star (c_k,0)\big) = \sum_{i=1}^{k} \Pi(c_i).$$

Proof

Induct on k. For $k=2$ this is the definition of $\star$. If the formula holds for k, then multiplying by $(c_{k+1},0)$ gives second coordinate

$$\sum_{i<j\le k} B(c_i,c_j) + B\left(\sum_{i=1}^{k} c_i, c_{k+1}\right) = \sum_{i<j\le k} B(c_i,c_j) + \sum_{i=1}^{k} B(c_i,c_{k+1}),$$

which is the pairwise sum for $k+1$ charges.

For the normal-ordered degree,

$$\Pi\left(\sum_i c_i\right) = \sum_i \Pi(c_i) + \sum_{i<j} B(c_i, c_j).$$

Subtracting the defect gives

$$\Pi = \sum_i \Pi(c_i).$$

∎

This is the clean first-principles construction: the BKM/Fock degree is the **normal-ordered Gram degree** of a physical extension flag.

7. Raw Π-descent is too strong

Now prove the obstruction to the swarm's proposed B-isotropic theorem.

Let

$$L = \bigoplus_{c \in \Gamma_X^{\mathrm{phys}}} L_c$$

be a physical-charge-graded Lie superalgebra with

$$[L_c, L_{c'}] \subseteq L_{c+c'}.$$

If one tries to push L forward using raw Π, one sets

$$(\Pi_* L)_T = \bigoplus_{\Pi(c)=T} L_c.$$

Proposition 7.1 — Raw Π-descent is equivalent to bracket B-isotropy

The raw pushforward bracket is Γ_{ind}-additive,

$$[(\Pi_* L)_T, (\Pi_* L)_{T'}] \subseteq (\Pi_* L)_{T+T'},$$

if and only if

$$[L_c, L_{c'}] \neq 0 \quad \Longrightarrow \quad B(c, c') = 0.$$

Proof

If $x \in L_c$, $y \in L_{c'}$, then

$$[x, y] \in L_{c+c'}.$$

Its raw Gram degree is

$$\Pi(c + c') = \Pi(c) + \Pi(c') + B(c, c').$$

For this to lie in degree

$$\Pi(c) + \Pi(c'),$$

one must have

$$B(c, c') = 0$$

whenever $[x, y] \neq 0$. Conversely, if this condition holds, every nonzero bracket has

$$\Pi(c + c') = \Pi(c) + \Pi(c'),$$

so the raw pushforward is Γ_{ind}-graded. ∎

So B-isotropy is not merely sufficient. It is necessary for raw Π-descent.

8. No-go theorem for raw B-isotropic descent to the full BKM algebra

The full $\mathfrak{g}_{\Delta_5}$ has real simple roots

$$\delta_1, \delta_2, \delta_3$$

with Cartan matrix

$$A = \begin{matrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{matrix}.$$

Thus for $i \neq j$,

$$(\operatorname{ad} e_i)^3 e_j = 0,$$

but the lower brackets

$$[e_i, e_j], \qquad [e_i, [e_i, e_j]]$$

are nonzero in the BKM algebra. Your manuscript already records the corresponding low-degree rows, including the $2\delta_i + \delta_j$ real-root rows with signed multiplicity 1.

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Theorem 8.1 — Raw B-isotropic Π-descent cannot realize the full real-root BKM strings

Suppose L is a physical-charge-graded Lie superalgebra whose raw Π-pushforward realizes the positive real-root part of $\mathfrak{g}_{\Delta_5}$. Let e_i, e_j be homogeneous lifts of two distinct even real simple root generators with physical charges c_i, c_j, and suppose

$$\Pi(c_i) = \gamma_i \neq 0, \qquad \Pi(c_j) = \gamma_j.$$

If

$$[e_i, e_j] \neq 0$$

and

$$[e_i, [e_i, e_j]] \neq 0,$$

then raw Π-descent is impossible.

Proof

Since $[e_i, e_j] \neq 0$, Proposition 7.1 forces

$$B(c_i, c_j) = 0.$$

The bracket $[e_i, e_j]$ has physical charge

$$c_i + c_j.$$

Since $[e_i, [e_i, e_j]] \neq 0$, Proposition 7.1 again forces

$$B(c_i, c_i + c_j) = 0.$$

But

$$B(c_i, c_i + c_j) = B(c_i, c_i) + B(c_i, c_j) = B(c_i, c_i).$$

Now

$$B(c_i, c_i) = \left(Q_i^2,\, 2Q_i \cdot P_i,\, P_i^2\right) = 2\Pi(c_i) = 2\gamma_i.$$

Because $\gamma_i \neq 0$, this is nonzero. Contradiction. ∎

So the proposed theorem

$$\text{“primitive subspace is } B\text{-isotropic, hence raw } \Pi\text{-descends to BKM”}$$

cannot be the full theorem. It can hold only for first-collision channels or for an abelian/short-root truncation. The full $\mathfrak{g}_{\Delta_5}$ needs the B-twisted normal-ordered extension.

This is a decisive structural point.

9. Explicit real-simple-root physical lifts

Although raw B-isotropy fails globally, the first simple collisions can be lifted cleanly.

Let Λ contain three orthogonal hyperbolic planes

$$U_1 \oplus U_2 \oplus U_3,$$

with bases e_i, f_i satisfying

$$e_i^2 = f_i^2 = 0, \qquad e_i \cdot f_i = 1.$$

The three type-II simple roots correspond to Gram triples

$$\gamma_1 = (1,1,0), \qquad \gamma_2 = (0,1,1), \qquad \gamma_3 = (0,-1,0).$$

Indeed,

$$\alpha(1,1,0) = \delta_1, \qquad \alpha(0,1,1) = \delta_2, \qquad \alpha(0,-1,0) = \delta_3.$$

Define physical charges

$$c_1 = (Q_1, P_1) = (e_1 + f_1, e_1),$$
$$c_2 = (Q_2, P_2) = (e_2, e_2 + f_2),$$
$$c_3 = (Q_3, P_3) = (e_3, -f_3).$$

Then

$$\Pi(c_1) = (1,1,0),$$

$$\Pi(c_2) = (0, 1, 1),$$

$$\Pi(c_3) = (0, -1, 0).$$

Because the three hyperbolic planes are orthogonal,

$$B(c_i, c_j) = 0 \qquad (i \neq j).$$

Thus the first brackets

$$[e_i, e_j]$$

can have raw additive Gram degree. But the second real-root brackets require defect:

$$[e_i, [e_i, e_j]]$$

has physical charge

$$2c_i + c_j.$$

Since

$$B(c_i, c_i) = 2\gamma_i,$$

one gets

$$\Pi(2c_i + c_j) = 4\gamma_i + \gamma_j,$$

whereas the BKM degree is

$$2\gamma_i + \gamma_j.$$

The missing term is exactly the defect

$$2\gamma_i = B(c_i, c_i).$$

In Γ_X,

$$(c_i, 0) \star (c_i, 0) \star (c_j, 0) = (2c_i + c_j, B(c_i, c_i)),$$

and therefore

$$\Pi\big((2c_i + c_j, B(c_i, c_i))\big) = \Pi(2c_i + c_j) - B(c_i, c_i) = 2\gamma_i + \gamma_j.$$

So the B-extension repairs the exact obstruction.

10. Corrected descent theorem

The right descent statement is this.

Definition 10.1 — B-normal-ordered primitive Hall algebra

A B-normal-ordered compact primitive Hall algebra is a Lie superalgebra

$$P_X = \bigoplus_{c \in \Gamma_X} (P_X)_c$$

such that

$$[(P_X)_c, (P_X)_{c'}] \subseteq (P_X)_{c \star c'}.$$

The Borcherds/lausa degree of a homogeneous element in degree c is

$$\Pi(c) \in \Gamma_{\mathrm{ind}}.$$

Define the normal-ordered pushforward

$$(\Pi_* P_X)_\gamma = \bigoplus_{\Pi(c)=\gamma} (P_X)_c.$$

Theorem 10.2 — Normal-ordered pushforward is automatically a Γ_{ind}-graded Lie superalgebra

If P_X is Γ_X-graded, then

$$[\Pi_* P_{X\,\gamma}, \Pi_* P_{X\,\gamma'}] \subseteq \Pi_* P_{X\,\gamma+\gamma'}.$$

Proof

Let

$$x \in (P_X)_c, \qquad y \in (P_X)_{c'}.$$

Then

$$[x, y] \in (P_X)_{c \star c'}.$$

Because Π is a group homomorphism,

$$\Pi(c \star c') = \Pi(c) + \Pi(c').$$

Therefore the bracket is additive in Γ_{ind}. ∎

This is the rigorous replacement for raw B-isotropic descent.

11. Conditional BKM recognition theorem, corrected

Now the actual candidate theorem should be stated as follows.

Theorem 11.1 — Conditional normal-ordered primitive recognition

Assume a compact oriented Hall/factorization realization of $X = K3 \times E$ supplies:

1. a B-normal-ordered primitive Lie superalgebra

$$P_X = H^0 \operatorname{Prim}^{\mathrm{prot}}(\mathcal{F}_X),$$

graded by

$$\Gamma_X = \Gamma_X^{\mathrm{phys}} \oplus_B \Gamma_{\mathrm{ind}};$$

2. a finite HN completion compatible with $\star$;
3. defect-zero elementary one-particle primitive lifts for the real and imaginary simple roots;
4. an oriented protected integration map;
5. a completed Hopf pairing and radical quotient;
6. after normal-ordered pushforward by Π, a primitive recognition certificate matching the Gritsenko–Nikulin Borcherds datum.

Then

$$\Pi_* P_X \cong \mathfrak{g}_{\Delta_5}$$

as Γ_{ind}-graded Lie superalgebras after the completed radical quotient.

Consequently,

$$\operatorname{sdim}(\Pi_* P_X)_{(n,l,m)} = f(nm, l)$$

on the active positive support, and its denominator is

$$\operatorname{den}(\Pi_* P_X) = 64^{-1} \Delta_5(2Z).$$

Proof

By Theorem 10.2, $\Pi_* P_X$ is a Γ_{ind}-graded Lie superalgebra.

The primitive recognition certificate supplies:

$$(\Pi_* P_X)_0 \simeq \Lambda_{II}^{2,1} \otimes \mathbb{C},$$

simple primitive representatives for the three real simple roots

$$\delta_1, \delta_2, \delta_3,$$

the Gritsenko–Nikulin imaginary simple-root fibres with parity, the Chevalley relations, the real Serre relations, the imaginary orthogonality relations, generation, PBW compatibility, and equality of the completed Hopf-pairing radical quotient.

Therefore the universal Borcherds–Kac presentation maps onto $\Pi_* P_X$. The no-extra-relations hypothesis identifies the kernel with the standard Borcherds–Serre/radical ideal. Thus

$$\Pi_* P_X \cong \mathfrak{g}_{\Delta_5}.$$

The denominator identity and signed multiplicity formula are then the Gritsenko–Nikulin denominator theorem, already imported in the manuscript. ∎

The geometric construction of P_X remains Open Problem D. But the descent mechanism is now rigorous and no longer blocked by the quadratic Gram defect.

12. D6–D2–D0 check: the OP variables are Gram variables

For a one-dimensional subscheme

$$Y \subset X = S \times E$$

with curve class

$$[Y] = (\beta, d)$$

and

$$\chi(\mathcal{O}_Y) = k,$$

the ideal sheaf satisfies

$$\mathrm{ch}(I_Y) = 1 - \mathrm{PD}[Y] - k[\mathrm{pt}_X].$$

Since

$$\mathrm{td}(S) = (1, 0, 1), \qquad \mathrm{td}(E) = 1,$$

one obtains

$$v_X(I_Y) = (1, 0, 1-d) \otimes 1_E + (0, -\beta, -k) \otimes \omega_E.$$

So define

$$P_Y = (1, 0, 1-d), \qquad Q_Y = (0, -\beta, -k).$$

Then

$$Q_Y^2 = \beta^2,$$

$$P_Y^2 = 2(d-1),$$

$$Q_Y \cdot P_Y = k.$$

If

$$\beta^2 = 2h - 2,$$

then

$$\Pi(Q_Y, P_Y) = (h-1, k, d-1).$$

So the OP/Borcherds monomial

$$q^{h-1} r^k s^{d-1}$$

is exactly

$$q^{Q^2/2} r^{Q \cdot P} s^{P^2/2}.$$

The scalar OP/DT theorem in the manuscript then says

$$Z^X_{\mathrm{OP/DT}} = -4096\, \Delta_5^{-2},$$

while the first-order determinant has

$$D_X = \Delta_5.$$

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13. Pfaffian/orientation theorem

Now make the Dirac/Pfaffian part precise.

For a compact Calabi–Yau threefold X, Joyce–Upmeier prove natural orientation data for moduli stacks of coherent sheaves and perfect complexes; in their formulation, orientation data is a square root of the natural canonical-type line bundle compatible with short exact sequences. Since $K3 \times E$ is a compact Calabi–Yau threefold, this gives the correct general existence input for an unreduced CY3 sheaf/perfect-complex orientation datum. arXiv

The Borisov–Joyce and Oh–Thomas theories are closely related Pfaffian/orientation technologies in the CY4 setting: Borisov–Joyce construct virtual classes for proper oriented (-2)-shifted symplectic derived schemes, and Oh–Thomas construct algebraic CY4 virtual cycles using localized square-root Euler classes. These are important analogues of the square-root/Pfaffian mechanism, but they do not by themselves prove the reduced $K3 \times E$ Hall orientation monodromy. arXiv +1

Theorem 13.1 — Orientation monodromy is a character

Assume a compact Hall/factorization realization supplies an orientation line

$$\mathfrak{o}$$

with

$$\mathfrak{o}^{\otimes 2} \simeq \mathscr{D},$$

where $\mathscr{D}$ is the determinant/virtual canonical line of the protected critical theory.

Assume a lifted action of the type-II Weyl group

$$W^{(2)}(\Lambda_{II}^{2,1})$$

on the oriented completed Hall object. For each $w \in W^{(2)}(\Lambda_{II}^{2,1})$, suppose the lift acts on $\mathfrak{o}$ by a sign

$$w^*\mathfrak{o} \simeq \epsilon_o(w)\mathfrak{o}, \qquad \epsilon_o(w) \in \{\pm 1\}.$$

Then

$$\epsilon_o : W^{(2)}(\Lambda_{II}^{2,1}) \to \{\pm 1\}$$

is a group character.

Proof

Coherence of the lifted action gives

$$(w_1 w_2)^* \mathfrak{o} \simeq w_2^*(w_1^* \mathfrak{o}).$$

If

$$w_1^* \mathfrak{o} \simeq \epsilon_o(w_1)\mathfrak{o}$$

and

$$w_2^* \mathfrak{o} \simeq \epsilon_o(w_2)\mathfrak{o},$$

then

$$(w_1 w_2)^* \mathfrak{o} \simeq \epsilon_o(w_1)\epsilon_o(w_2)\mathfrak{o}.$$

Thus

$$\epsilon_o(w_1 w_2) = \epsilon_o(w_1)\epsilon_o(w_2).$$

∎

Theorem 13.2 — Simple-reflection signs imply the Maass character on the type-II Weyl group

Assume

$$\epsilon_o(s_{\delta_i}) = -1, \qquad i = 1, 2, 3,$$

for the three type-II simple reflections.

Then

$$\epsilon_o = \det$$

on

$$W^{(2)}(\Lambda_{II}^{2,1}).$$

Since the manuscript proves

$$\nu_{\Delta_5}|_{W^{(2)}(\Lambda_{II}^{2,1})} = \det,$$

one obtains

$$\epsilon_o = \nu_{\Delta_5}$$

on the type-II Weyl group.

Proof

The group $W^{(2)}(\Lambda_{II}^{2,1})$ is generated by the three simple reflections

$$s_{\delta_1}, s_{\delta_2}, s_{\delta_3}.$$

Both ϵ_o and $\det$ are characters. On every generator,

$$\epsilon_o(s_{\delta_i}) = -1 = \det(s_{\delta_i}).$$

Therefore the two characters agree on the generating set, hence agree on the whole group.

The equality with ν_{Δ_5} follows from the manuscript's multiplier restriction computation. ∎

So the orientation theorem is reduced to the following concrete geometric calculation:

$$\text{compute the determinant-line monodromy around the three simple type-II walls a}$$

That is the genuinely close orientation theorem.

14. Pfaffian determinant theorem

Assume the corrected normal-ordered primitive recognition theorem holds, so that

$$\Pi_* P_X \cong \mathfrak{g}_{\Delta_5}.$$

Define the protected Pfaffian determinant by

$$\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X)(Z) = 64 q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{\mathrm{sdim}(\Pi_* P_X)_{(n,l,m)}}.$$

If

$$\mathrm{sdim}(\Pi_* P_X)_{(n,l,m)} = f(nm, l),$$

then

$$\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X)(Z) = \Delta_5(Z).$$

Proof

Substitute the signed multiplicities:

$$\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X)(Z) = 64 q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)}.$$

By the Borcherds–Gritsenko–Nikulin product formula imported in the manuscript, this product is

$$\Delta_5(Z).$$

∎

The scalar square is then

$$Z^X_{\mathrm{OP/DT}} = -4096\ \mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X)^{-2}.$$

So the OP/DT theory is the scalar, orientation-forgetting square; the protected Pfaffian is the first-order object.

15. Final corrected theorem package

The strongest rigorous manuscript-level upgrade is:

Theorem package

Let

$$\Gamma_X^{\mathrm{phys}} = \Lambda \oplus \Lambda, \qquad \Pi(Q,P) = \left(Q^2/2, Q\cdot P, P^2/2\right).$$

Then:

1. Π is surjective onto Γ_{ind} when Λ contains $U \oplus U$, and every (n,l,m) has an explicit primitive lift.
2. The failure of Π to be additive is the symmetric bilinear cocycle

$$B(c,c') = (Q\cdot Q', Q\cdot P' + Q'\cdot P, P\cdot P').$$

3. The B-twisted charge extension

$$\Gamma_X = \Gamma_X^{\mathrm{phys}} \oplus_B \Gamma_{\mathrm{ind}}$$

is associative and abelian.

4. The normal-ordered Gram map

$$\Pi(c, \kappa) = \Pi(c) - \kappa$$

is additive.

5. Raw B-isotropic Π-descent cannot realize the full $\mathfrak{g}_{\Delta_5}$, because the real-root strings require nonzero brackets

$$[e_i, [e_i, e_j]] \neq 0,$$

but raw B-isotropy would force

$$B(c_i, c_i) = 0,$$

contradicting

$$B(c_i, c_i) = 2\Pi(c_i) \neq 0.$$

6. The corrected compact theorem is therefore normal-ordered descent:

$$H^0 \operatorname{Prim}^{\mathrm{prot}}(\mathcal{F}_X) \text{ should be } \Gamma_X\text{-graded, and } \Pi_* H^0 \operatorname{Prim}^{\mathrm{prot}}(\mathcal{F}_X) \cong \mathfrak{g}_{\Delta_5}.$$

7. If the orientation line exists, descends through the reduced/hybrid Hall construction, and has sign -1 on the three simple type-II reflections, then

$$\epsilon_o = \nu_{\Delta_5}$$

on the type-II Weyl group.

8. Under the same recognition data,

$$\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X) = \Delta_5,$$

and the scalar OP/DT branch is

$$Z^X_{\mathrm{OP/DT}} = -4096\, \Delta_5^{-2}.$$

This is the constructible Dirac-inspired formulation: the square root is not raw Π-descent. It is a **normal-ordered first-order algebra** on the B-twisted charge extension, whose Pfaffian determinant is Δ_5.

Sources

Sources

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PDF

here's the latest version of my WIP whitepaper. Attack it from all angles, and as before, suggest exactly how to level it up into the strongest version of this whitepaper conceivable

Show more

I'll treat this as a hostile referee report plus construction blueprint: find what breaks, extract theorem-grade mathematics, and upgrade the Dirac/Pfaffian, charge-descent, locality, and boundary frameworks.

The draft is now stronger on charge descent and locality; the main new attack is that conditional recognition theorems may conceal the missing constructions as hypotheses.

Thought for 8m 12s

According to the April 28, 2026 draft, the paper has become much more serious. It now has the right ambition: **Dirac first-order object → Pfaffian square root → scalar square**, with the compact $K3 \times E$ Hall/factorization construction treated as the real open problem rather than hidden behind the automorphic product. That is the right direction. The strongest version of this paper is not a smaller paper. It is a paper that makes three things brutally precise:

$$\textbf{automorphic theorem} \quad + \quad \textbf{normal-ordered physical charge descent} \quad +$$

Right now the draft is close, but it still contains several places where conditional recognition is written as if it were construction. Those are the places a serious referee will attack.

The latest draft's strongest achievement is the introduction of the additive physical charge lattice Γ_X^{phys}, the non-additive Gram map Π_X, the bilinear defect cocycle B, and the cocycle-corrected descent target $\Pi_{X,*}^{\Theta}$. That is exactly the structural repair the paper needed. The draft now explicitly states that the upstairs primitive bracket is physically graded by Γ_X^{phys}, while the BKM comparison happens only after cocycle-corrected Gram descent. raeez.lorgat.automorphic-correc...

But the paper is still vulnerable in four major zones:

$$\text{orientation, equivariant descent, normal-ordering, and factorization locality.}$$

Below is the attack and the upgrade path.

1. The central theorem should not be "Pfaffian recognition"; it should be "Dirac–Igusa conditional theorem"

The draft's Dirac thesis is excellent:

$$\Delta_5 \quad \text{is not an analytic branch of} \quad \Delta_5^2.$$

It is supposed to be the Pfaffian of a first-order protected object $\mathfrak{D}_X$, whose scalar orientation-forgetting square gives

$$Z_{\mathrm{OP/DT}}^X = -4096\,\Delta_5^{-2}.$$

The abstract now states this cleanly: the OP/DT scalar branch is orientation-forgetful, while the recognition target is a first-order protected operator/algebra $\mathfrak{D}_X$ with protected Pfaffian Δ_5. raeez.lorgat.automorphic-correc...

The problem is Theorem 3.10. It assumes orientation data $(O1)$, Weyl-equivariant cocycle extension $(O1)^+$, and a Pfaffian local model $(O2)$, then concludes that the first-order protected operator/algebra $\mathfrak{D}_X$ exists. But orientation data alone does **not** construct:

$$\mathfrak{D}_X,$$

nor a chiral protected module, nor a spectrum, nor compact primitive states, nor the mode decomposition whose Pfaffian has exponents $f(nm, l)$. The proof currently defines $\mathfrak{D}_X$ by saying its Berezinian/Pfaffian on the charge component is the canonical Pfaffian and then collapses the product to $v_0 \operatorname{sdet} V(1-x)$. That is not a construction

of $\mathfrak{D}_X$; it is a **renaming of the already-known K_0-determinant after assuming the missing first-order object exists.** raeez.lorgat.automorphic-correc...

External orientation theory supports the existence of orientation data for moduli of coherent sheaves/perfect complexes on compact Calabi–Yau threefolds, but that is not the same as constructing the reduced E-quotiented, Weyl-equivariant, compact Hall/factorization object needed here. Joyce–Upmeier prove natural orientation data for compact Calabi–Yau threefolds, i.e. square-root line bundles satisfying compatibility over short exact sequences, but your reduced $K3 \times E$ quotient plus Weyl action is an extra layer. arXiv

Fix

Replace Theorem 3.10 with two separate statements.

Theorem A: Pfaffian sign theorem

Assume a compact Dirac–Igusa datum exists:

$$\mathfrak{D}_X, \quad \mathcal{H}_X^{\mathrm{prot}}, \quad o_C^{\otimes 2} \simeq \det \mathrm{RHom}(C, C), \quad \Pi_{X,*}^{\Theta},$$

and assume its protected one-particle superdimensions are

$$\mathrm{sdim}\, P_{n,l,m}^{\Pi} = f(nm, l).$$

Then

$$\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X) = \Delta_5.$$

Theorem B: Orientation-character theorem

Assume the orientation monodromy character ϵ_o exists and satisfies

$$\epsilon_o(s_{\delta_i}) = -1, \qquad i = 1, 2, 3.$$

Then

$$\epsilon_o = \nu_{\Delta_5} \quad \text{on} \quad W^{(2)}(\Lambda_{II}^{2,1}).$$

This second theorem is genuinely close and mathematically clean. The first is a conditional consequence of the missing compact object, not a construction of it.

2. The Mukai-vector lemma must be corrected immediately

Lemma 5.1 is one of the most important places in the paper because it connects the OP/DT variables to the physical Gram variables. But it currently says:

> "On a smooth projective fourfold..."

This is wrong. $K3 \times E$ is a Calabi–Yau **threefold**, not a fourfold. The same lemma also leaves

$$k = n - (\text{Todd correction})$$

unspecified. That weakens one of the paper's core physical derivations.

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You should make the result exact.

Let $Y \subset X = S \times E$ be one-dimensional with

$$[Y] = (\beta, d), \qquad \chi(\mathcal{O}_Y) = n.$$

Then

$$\mathrm{ch}(I_Y) = 1 - \mathrm{ch}(\mathcal{O}_Y).$$

For a one-dimensional sheaf on a Calabi–Yau threefold,

$$\mathrm{ch}_2(\mathcal{O}_Y) = \mathrm{PD}[Y], \qquad \mathrm{ch}_3(\mathcal{O}_Y) = \chi(\mathcal{O}_Y) = n.$$

Since

$$\mathrm{td}(S) = (1, 0, 1), \qquad \mathrm{td}(E) = 1,$$

one obtains

$$v_X(I_Y) = (1, 0, 1 - d) \otimes 1_E + (0, -\beta, -n) \otimes \omega_E.$$

Thus the correct electric-magnetic pair is

$$P_Y = (1, 0, 1 - d), \qquad Q_Y = (0, -\beta, -n).$$

Then

$$\frac{Q_Y^2}{2} = \frac{\beta^2}{2} = h - 1,$$

$$Q_Y \cdot P_Y = n,$$

$$\frac{P_Y^2}{2} = d - 1.$$

Therefore

$$\Pi(Q_Y, P_Y) = (h - 1, n, d - 1).$$

This is stronger than the current statement. There is no vague Todd correction if $n = \chi(\mathcal{O}_Y)$. If your n is not $\chi(\mathcal{O}_Y)$, rename the variables so the correction is explicit.

This corrected lemma should be one of the paper's central theorems, not a side lemma.

3. The E-equivariant descent lemma is currently wrong

Lemma 5.8 is the most dangerous mathematical section in the draft. It says the equivariant Serre spectral sequence has

$$H^1(BE; \mathbb{Z}/2) \cong (\mathbb{Z}/2)^2$$

and then writes

$$H^2(BE; \mathbb{Z}/2) = \mathbb{Z}/2\langle \alpha\beta \rangle \oplus \mathbb{Z}/2\langle \alpha^2 \rangle \oplus \mathbb{Z}/2\langle \beta^2 \rangle.$$

That is the cohomology pattern of $B(\mathbb{Z}/2)^2$, not of BE for $E^{\mathrm{top}} \cong T^2$. For a connected torus,

$$BE \simeq BT^2 \simeq K(\mathbb{Z}^2, 2),$$

so

$$H^*(BE; \mathbb{F}_2) = \mathbb{F}_2[u_1, u_2], \qquad |u_i| = 2.$$

Hence

$$H^1(BE; \mathbb{F}_2) = 0,$$

and

$$H^2(BE; \mathbb{F}_2) \cong (\mathbb{F}_2)^2,$$

not $(\mathbb{F}_2)^3$. The $(\mathbb{F}_2)^3$ degree-two group belongs to

$$H^2(BE[2]; \mathbb{F}_2), \qquad E[2] \cong (\mathbb{Z}/2)^2,$$

which is a different object. The draft later correctly treats $E[2]$ as Klein-four, but the free connected E-quotient and the finite stabilizer $E[2]$-quotient are being mixed.

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Fix

Split Lemma 5.8 into two independent results.

Lemma 5.8A: Free connected E-descent

For the free E-locus,

$$E_2^{p,q} = H^p(BE; H^q(\mathfrak{M}_C; \mathbb{F}_2)), \qquad H^*(BE; \mathbb{F}_2) = \mathbb{F}_2[u_1, u_2], \quad |u_i| = 2.$$

There is no $E_2^{1,1}$ obstruction from $H^1(BE)$, because $H^1(BE) = 0$. The first connected equivariant obstruction lives in total degree two through

$$E_2^{2,0} = H^2(BE; \mathbb{F}_2) \otimes H^0(\mathfrak{M}_C; \mathbb{F}_2),$$

and in

$$E_2^{0,2} = H^0(BE; H^2(\mathfrak{M}_C; \mathbb{F}_2)).$$

So the descent obstruction must be rewritten as a class in

$$H_E^2(\mathfrak{M}_C; \mathbb{F}_2),$$

with connected-torus part lying in $\mathbb{F}_2 u_1 \oplus \mathbb{F}_2 u_2$, not in $\alpha^2, \alpha\beta, \beta^2$.

Lemma 5.8B: Finite stabilizer $E[N]$-descent

On small-orbit strata, the stabilizer is finite:

$$E[N] \cong (\mathbb{Z}/N)^2.$$

For $N = 2$,

$$E[2] \cong (\mathbb{Z}/2)^2,$$

and then

$$H^*(BE[2]; \mathbb{F}_2) = \mathbb{F}_2[x_1, x_2], \qquad |x_i| = 1,$$

so

$$H^2(BE[2]; \mathbb{F}_2) = \mathbb{F}_2\langle x_1^2, x_1x_2, x_2^2\rangle.$$

This is where the rank-three obstruction group belongs.

This correction is non-negotiable. Leaving the current version will invite an immediate referee rejection.

4. The small-orbit gerbe vanishing is not proved

Lemma 5.9 says the $E[2]$-gerbe residual vanishes because translation acts trivially on $\mathrm{Pic}(E)$, on $H^*(E)$, and on the trivial cotangent bundle. raeez.lorgat.automorphic-correc...

That is not enough.

A group can act trivially on the underlying line bundle while acting nontrivially on its **linearization**. The obstruction lives exactly in the difference between trivial underlying line and nontrivial equivariant structure. Translation acting trivially on Ω^1_E does not automatically trivialize the orientation gerbe on the moduli stack. You must compute the equivariant determinant line, not only the ordinary determinant line.

The claim

$$\beta_C^{E,E[2]} = 0$$

may be true on a rank-one stratum, but the present proof does not establish it.

Fix

Replace Lemma 5.9 by a conditional criterion:

> If the $E[2]$-linearization of the reduced determinant line is the trivial character on each of the three nontrivial cyclic subgroups, then the $E[2]$-gerbe class vanishes.

Then separately propose a computation:

$$\chi_{o_C} : E[2] \to \{\pm 1\}$$

from the equivariant determinant of

$$\mathrm{RHom}_X(I_Y, I_Y)_{\mathrm{red}}.$$

The actual theorem should be:

$$\beta_C^{E,E[2]} = 0 \iff \chi_{o_C} = 1 \quad \text{and the associated } H^2(BE[2];\mathbb{F}_2) \text{ class vanishes.}$$

That would be rigorous and checkable.

5. The OP minus sign must not be turned into orientation data

The draft correctly says in several places that 64, 4096, and the OP minus sign are normalization data, not orientation data. But then it says the cosection-induced parity flip "sources the $(-1)^F$ sign" and "is the sign that makes"

$$Z^X_{\mathrm{OP}} = -4096\Delta_5^{-2}$$

carry its overall minus. raeez.lorgat.automorphic-correc...

That is dangerous. It blurs the exact distinction you are trying to enforce.

The OP scalar sign is a convention in the scalar enumerative branch. The Hall/Pfaffian orientation character is a monodromy character:

$$\epsilon_o : W^{(2)}(\Lambda^{2,1}_{II}) \to \{\pm 1\}.$$

These are different kinds of signs. The OP minus is a scalar prefactor. The Maass character is a transformation character. A scalar prefactor cannot determine a character.

Fix

Write:

$$Z^X_{\mathrm{OP/DT}} = -4096\,\Delta_5^{-2}$$

and then say:

> The leading minus is a scalar convention of the OP/DT branch. It may be compatible with a reduced parity convention, but it is not evidence for the reflection character ϵ_o. The reflection character must be computed from orientation-line monodromy around the type-II walls.

Never say the OP minus "sources" the orientation character.

6. The B-cocycle material is strong, but the descent should be normal-ordered rather than "shifted away"

The charge-lattice upgrade is the best new mathematics in the draft. You now define

$$\Gamma_X^{\mathrm{phys}} = \Lambda_S \oplus \Lambda_S,$$

$$\Pi_X(Q,P) = \left(Q^2/2, Q\cdot P, P^2/2\right),$$

and

$$B(c,c') = (Q\cdot Q',\, Q\cdot P' + Q'\cdot P,\, P\cdot P').$$

The draft proves

$$\Pi_X(c+c') = \Pi_X(c) + \Pi_X(c') + B(c,c'),$$

and the central product

$$(c,T)\star(c',T') = (c+c', T+T'+B(c,c')).$$

This is correct and important. raeez.lorgat.automorphic-correc...

But the language "subtracting $B(c,c')$" by a cochain Θ can sound like you are forcing non-additive physics into additive Gram degrees by fiat. The safer and stronger construction is **normal-ordering**.

Define

$$\Gamma_X = \Gamma_X^{\mathrm{phys}} \oplus_B \Gamma_{\mathrm{gram}}.$$

Then define the additive normal-ordered Gram map

$$\Pi : \Gamma_X \to \Gamma_{\mathrm{gram}}, \qquad \Pi(c,T) = \Pi(c) - T.$$

Because

$$\Pi(c+c') = \Pi(c) + \Pi(c') + B(c,c'),$$

one has

$$\Pi\big((c,T)\star(c',T')\big) = \Pi(c,T) + \Pi(c',T').$$

So the correct theorem is:

> If the protected primitive Hall algebra is Γ_X-graded and its bracket respects $\star$, then its normal-ordered pushforward by Π is a Γ_{gram}-graded Lie superalgebra.

This avoids having to assert that a mysterious degree-shift isomorphism exists. The Θ-formalism can remain, but it should be presented as one possible categorified implementation of normal-ordering, not as a proven operation.

The draft already has a conditional Θ-descent theorem and says it does not erase B; it transports the raw target to the additive Gram target. raeez.lorgat.automorphic-correc... That is conceptually right. Make it mathematically cleaner by putting Γ_X and Π first, then Θ as a categorification.

7. Add a no-go theorem for raw Π-descent

This would be a major level-up.

The paper should prove that raw Π-descent cannot realize the full BKM bracket. That makes your cocycle-corrected descent not optional but forced.

Let $P = \bigoplus_c P_c$ be physically graded:

$$[P_c, P_{c'}] \subset P_{c+c'}.$$

Raw Π-descent would require

$$\Pi(c + c') = \Pi(c) + \Pi(c')$$

for every nonzero bracket. Therefore it requires

$$B(c, c') = 0$$

for every nonzero bracket.

But in the BKM algebra, for distinct real simple roots e_i, e_j, one has

$$[e_i, e_j] \neq 0, \qquad [e_i, [e_i, e_j]] \neq 0,$$

while

$$(\operatorname{ad} e_i)^3 e_j = 0.$$

If c_i is a physical lift of e_i, then the second bracket forces

$$B(c_i, c_i + c_j) = 0.$$

If also the first bracket forced $B(c_i, c_j) = 0$, then

$$B(c_i, c_i + c_j) = B(c_i, c_i) + B(c_i, c_j) = B(c_i, c_i).$$

But

$$B(c_i, c_i) = 2\Pi(c_i) \neq 0.$$

Contradiction.

Therefore:

$$\text{raw } \Pi\text{-descent cannot realize the full real-root BKM strings.}$$

This theorem would be genuinely valuable. It proves that the Dirac/normal-ordering data is structurally necessary.

8. The local factorization story is physically right but mathematically underdefined

The paper now states the local object comes first, that protected local observables should factorize before any trace is taken, and that positive elliptic degree is wrapped/global over E. This is one of the strongest physical insights in the draft.

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The issue is that "holomorphic E_3-factorization algebra" remains too ambiguous.

PTVV gives the shifted-symplectic background: moduli of perfect complexes on a d-Calabi–Yau carry $(2-d)$-shifted symplectic structures, hence for a Calabi–Yau threefold one expects (-1)-shifted symplectic geometry and d-critical/vanishing-cycle structures. arXiv

But "holomorphic E_3" is not automatically the same as ordinary real little-disks E_3. A holomorphic theory on a complex threefold gives holomorphic factorization, and only after a specified formality/framing/local model does one get an E_3-algebraic shadow.

Fix

Define the structure in three layers:

$$\mathcal{A}_X \in \mathrm{Fact}^{\mathrm{hol}}(X; \mathrm{Ch}_{\mathbb{C}})$$

first.

Then define the local pointwise model:

$$\mathcal{A}^{\mathrm{loc}}_{X,x} \in E_3\text{-}\mathrm{Alg}$$

only after choosing:

1. a holomorphic disk/polydisk model near x,
2. a compact-support condition,
3. a formality quasi-isomorphism,
4. a framing/homotopy datum,
5. QME/anomaly cancellation.

Then define the $K3 \to E$ specialization:

$$\mathrm{Sp}^{\mathrm{ch}}_{K3,E}(\mathcal{A}_X)$$

as an actual functor or correspondence, not just a named placeholder.

The draft's Definition 10.29 moves in this direction by listing $A_X^{E_3}$, $Sp^{ch}_{K3,E}$, F_X, H_X, Θ_Π, Θ_{Kos}, and Π^{rec}. raeez.lorgat.automorphic-correc... But for the strongest version, you should split this into:

$$\text{local BV construction}, \quad \text{charge completion}, \quad \text{K3 integration/specialization}, \quad \text{hyb}$$

Each gets its own theorem or open problem.

9. The hybrid wrapped sector should be promoted to a real mathematical object

The draft now says the full s-direction requires hybrid local/global wrapped factorization, not naive $\mathrm{Ran}(E)$-locality. That is correct. The current Open Problem 10.15 is a good start. raeez.lorgat.automorphic-correc...

But to make this groundbreaking, define a new object.

Call it:

$$\mathrm{Fact}^{\mathrm{hyb}}(E; \mathbb{N}_{\mathrm{wrap}})$$

or

$$\mathrm{Ran}^{\mathrm{hyb}}(E).$$

Objects should be pairs

$$(F, b),$$

where $F \subset E$ is finite support and $b \geq 0$ is elliptic wrapping degree. The $b = 0$ stratum is ordinary Ran locality. The $b > 0$ strata are global wrapped sectors.

Then define

$$\mathcal{F}_X^{\mathrm{hyb}}(F, b)$$

with:

$$\mathcal{F}_X^{\mathrm{hyb}}(F, 0) = \text{projection-finite local sector},$$

$$\mathcal{F}_X^{\mathrm{hyb}}(F, b > 0) = \text{wrapped module sector}.$$

The structure maps should be:

$$\mathcal{F}(F_1, b_1) \otimes \cdots \otimes \mathcal{F}(F_k, b_k) \to \mathcal{F}(F_1 \cup \cdots \cup F_k, b_1 + \cdots + b_k + \text{interaction}),$$

where the interaction is measured upstairs in Γ_X^{phys}, then normal-ordered through Π.

This would turn the locality obstruction into a new mathematical structure. That is exactly the kind of synthesis that could advance knowledge.

10. The physical charge lattice should be explicitly distinguished from the full $N = 4$ charge lattice

The draft uses

$$\Gamma_X^{\mathrm{phys}} = H(S, \mathbb{Z}) \oplus H(S, \mathbb{Z}).$$

This is correct for the algebraic even Mukai sector. But strong physics demands a statement about the full four-dimensional $N = 4$ charge lattice.

For type II on $K3 \times T^2$, the full electric/magnetic charge lattice is usually described in terms of a Narain-type lattice, with electric and magnetic charges living in a larger

lattice than the algebraic Mukai sublattice. The Igusa variables are the T-duality invariants

$$(Q^2/2,\ Q\cdot P,\ P^2/2),$$

but the full microscopic charge includes more data than the Gram triple. The D6–D2–D0 algebraic sector is a geometric sublocus.

Fix

Add a paragraph:

> In the full $N=4$ compactification the physical charge lattice is larger; the paper works in the algebraic even Mukai sector relevant to the D6–D2–D0/OP branch. The map Π_X is the restriction of the physical electric-magnetic Gram map to this sector.

This prevents a physicist from accusing the paper of confusing the full U-duality charge lattice with the algebraic D-brane sublattice.

11. The orientation-character proof should be stripped to its clean core

The current Lemma 3.11 contains a long “three-wall sign cocycle” and a claimed independent Maass-character verification. This is too complicated and partly circular.

The clean theorem is:

$$W^{(2)}(\Lambda_{II}^{2,1}) = \langle s_{\delta_1}, s_{\delta_2}, s_{\delta_3} \mid s_{\delta_i}^2 = 1\rangle$$

with no finite braid relations because the off-diagonal Cartan entries give infinite Coxeter exponents. Therefore its abelianization is

$$(\mathbb{Z}/2)^3.$$

Any character is determined by its signs on the three simple reflections.

If the Pfaffian local model gives

$$\epsilon_o(s_{\delta_i}) = -1$$

for all three simple type-II reflections, then

$$\epsilon_o = \det$$

on the type-II Weyl group.

The draft already has the Maass multiplier values:

$$\nu_{\Delta_5}(s_{\delta_i}) = -1,$$

and $+1$ on the chamber-permuting S_3 reflections. raeez.lorgat.automorphic-correc...

So the theorem is simply:

$$\epsilon_o(s_{\delta_i}) = -1 \quad \forall i \implies \epsilon_o = \nu_{\Delta_5} \text{ on } W^{(2)}(\Lambda_{II}^{2,1}).$$

The hard geometric problem is only the local Pfaffian computation

$$\epsilon_o(s_{\delta_i}) = -1.$$

Everything else is group theory.

Remove the three-wall cochain material. It creates more risk than value.

12. The “Bott-element sign normalization” should be removed

Remark 4.4 connects the identity

$$B(c, c) = 2\Pi_X(c)$$

to Bott-periodicity sign conventions. That is unnecessary and likely confusing. The identity

$$B(c, c) = 2\Pi_X(c)$$

is a purely lattice-theoretic polarization identity. It has nothing to do with Bott elements, odd parity sectors, or determinant-line squaring.

This remark makes readers think the paper is importing machinery to prove an elementary identity, and worse, that parity choices could change the Gram polarization. They cannot. The lattice identity is fixed.

Fix

Delete the Bott discussion from the Gram-cocycle section. Put determinant-line sign conventions only in the orientation section.

13. The scalar square section is good but should be even more explicit about what OP proves

The scalar branch is correctly imported:

$$Z^{X}_{\mathrm{OP/DT}} = -4096\Delta_5^{-2}.$$

Oberdieck–Pixton solve the reduced Gromov–Witten theory of $S \times E$ for primitive K3 classes and deduce the Igusa cusp form conjecture. arXiv

The draft correctly states that OP/DT gives a scalar Behrend-weighted quotient integration, not a Hall product, CoHA integration map, or orientation character. raeez.lorgat.automorphic-correc...

The improvement is to put the scalar theorem behind the Dirac principle:

$$\text{scalar} = \text{orientation-forgetful}\,.$$

Then state:

$$\mathrm{OP/DT}\ \text{proves the scalar square, not the first-order square root.}$$

This should become a boxed slogan.

14. The boundary-row part is useful but should be decoupled

The eight-row certificate ledger is valuable. It prevents false physical row maps. Cléry–Gritsenko prove exactly eight genus-two forms with simplest diagonal divisor. arXiv Govindarajan proves that the square roots of the CHL Siegel forms are dd-modular forms and arise as Weyl–Kac–Borcherds denominator formulae. arXiv

The draft's certificate framework is right: product row, denominator row, CHL scalar square, reduced CY/DT scalar host, and compact Hall/chiral host are separate structures. raeez.lorgat.automorphic-correc...

But in the strongest $N = 1$ Igusa paper, this material may distract from the main theorem. The reader will audit the row table instead of focusing on the Dirac–Igusa construction.

Fix

Keep the certificate idea, but make Part II explicitly independent:

> Part II is a boundary-row ledger. It is not used in the proof of the $N = 1$ theorem.

Or split it into a companion paper.

15. The spin L-factor appendix is probably still a liability

Appendix A is not structurally needed. It does not help construct:

$$A_X^{E_3}, \quad \Pi_{X,*}^{\Theta}, \quad \epsilon_o, \quad \mathfrak{D}_X, \quad \text{or} \quad \mathcal{F}_X^{\text{hyb}}.$$

It may be correct, but it creates another place for a referee to find a normalization issue unrelated to the thesis.

Fix

Move it to a separate "arithmetic normalization note," or reduce it to one sentence:

> The scalar square Δ_5^2 lies on a Saito–Kurokawa eigenline; this arithmetic fact is not used in the Dirac/Pfaffian construction.

16. The strongest theorem package

The strongest conceivable version of the whitepaper should have this theorem spine.

Theorem 1: Borcherds determinant theorem

$$D_X(Z) = 64 q^{1/2} r^{1/2} s^{1/2} \prod_{\Gamma_{\text{eff}}} (1 - q^n r^l s^m)^{f(nm,l)} = \Delta_5(Z).$$

This is the theta-normalized Borcherds–Gritsenko–Nikulin product. Gritsenko–Nikulin's automorphic correction theorem supplies the BKM denominator algebra and the Δ_5 product/denominator data. arXiv

Theorem 2: D6–D2–D0 Mukai-Gram theorem

For $Y \subset S \times E$ with $[Y] = (\beta_h, d)$, $\chi(\mathcal{O}_Y) = n$, and $\beta_h^2 = 2h - 2$,

$$v_X(I_Y) = (1, 0, 1 - d) \otimes 1_E + (0, -\beta_h, -n) \otimes \omega_E,$$

so

$$\Pi(Q_Y, P_Y) = (h - 1, n, d - 1).$$

This makes the OP variables physically natural.

Theorem 3: Gram cocycle theorem

$$B(c, c') = (Q \cdot Q', Q \cdot P' + Q' \cdot P, P \cdot P')$$

is symmetric bilinear, satisfies

$$\Pi(c + c') = \Pi(c) + \Pi(c') + B(c, c'),$$

and defines

$$\Gamma_X = \Gamma_X^{\text{phys}} \oplus_B \Gamma_{\text{gram}}.$$

Theorem 4: Normal-ordered descent theorem

Define

$$\Pi(c, T) = \Pi(c) - T.$$

Then

$$\Pi : \Gamma_X \to \Gamma_{\text{gram}}$$

is a homomorphism. Therefore any Γ_X-graded primitive Hall Lie superalgebra has a Γ_{gram}-graded normal-ordered pushforward.

Theorem 5: No-go theorem for raw Gram descent

Raw Π-descent cannot realize the full BKM real-root strings. Therefore the B-corrected/normal-ordered descent is necessary, not decorative.

Theorem 6: Scalar square theorem

$$Z_{\text{OP/DT}}^X = -4096\Delta_5^{-2}.$$

This is imported from OP/Oberdieck and is orientation-forgetful.

Theorem 7: Conditional Dirac–Igusa theorem

Assume a compact Dirac–Igusa datum:

$$\mathfrak{D}_X, \quad \mathcal{F}_X^{\mathrm{hyb}}, \quad \Gamma_X, \quad \Pi, \quad o_C, \quad \epsilon_o, \quad \Pi^{rec}.$$

If

$$H^0\Pi_* \operatorname{Prim}^{\mathrm{prot}}(\mathcal{F}_X^{\mathrm{hyb}}) \cong \mathfrak{g}_{\Delta_5},$$

then

$$\operatorname{Pf}_{\mathrm{prot}}(\mathfrak{D}_X) = \Delta_5,$$

and

$$Z^X_{\mathrm{OP/DT}} = -4096 \operatorname{Pf}_{\mathrm{prot}}(\mathfrak{D}_X)^{-2}.$$

Theorem 8: Orientation-character theorem

If the compact orientation monodromy exists and

$$\epsilon_o(s_{\delta_i}) = -1, \qquad i = 1, 2, 3,$$

then

$$\epsilon_o = \nu_{\Delta_5}$$

on $W^{(2)}(\Lambda^{2,1}_{II})$.

This theorem is clean, provable, and almost ready.

17. Exact edits I would make

Rename:

$$\text{“Theorem 3.10 Dirac/Pfaffian recognition under orientation existence”}$$

to:

$$\text{“Conditional Dirac–Pfaffian sign and normalization theorem.”}$$

Do not let it claim existence of $\mathfrak{D}_X$ from orientation alone.

Replace Lemma 5.1 with the exact CY3 Mukai-vector computation and remove “fourfold” and “Todd correction.”

Replace Lemma 5.8 with two lemmas: connected E-descent and finite $E[N]$-stabilizer descent.

Downgrade Lemma 5.9 to a conditional equivariant-linearization criterion.

Delete the Bott-element sign normalization remark.

Add the raw-Π-descent no-go theorem.

Make Γ_X and Π the primary descent mechanism; treat Θ as a categorified realization.

Define the hybrid wrapped factorization category/object instead of only describing it.

State explicitly that $H(K3) \oplus H(K3)$ is the algebraic even Mukai sector, not the full $N = 4$ charge lattice.

Move the L-factor appendix out of the main paper.

Bottom line

This is now a real research program. The paper’s possible breakthrough is not merely

$$D_X = \Delta_5.$$

That is beautiful but imported through Borcherds/Gritsenko–Nikulin.

The possible breakthrough is:

$$\text{a Dirac/Pfaffian compact protected algebra whose normal-ordered Gram descent i}$$

To make the paper maximally strong, the central new contribution should be:

$$\Gamma_X^{\text{phys}} \longrightarrow \Gamma_X \xrightarrow{\Pi} \Gamma_{\text{gram}} \longrightarrow \mathfrak{g}_{\Delta_5}$$

together with the proof that raw Π-descent is impossible and the formulation of hybrid wrapped factorization as the physical locality mechanism.

That would make the paper more than a careful normalization ledger. It would identify the exact new mathematical object demanded by the physics:

$$\text{a normal-ordered, orientation-sensitive, hybrid local/global Hall factorization alge}$$

That is the strongest version of the whitepaper.

Sources

Sources

generate an EXHAUSTIVE set of recommendations for technical fixes, modifications, new insertions, rewrites etc.

I'll treat this as a referee-grade technical refactor: isolate structural failure modes, then give section-by-section fixes, proof insertions, theorem hygiene, physics framing, and replacement text.

I'll target the April 2026 draft's current spine: Dirac framing, local observables, physical charge lattice, cocycle descent, and orientation. Key fixes include Mukai vectors, BE cohomology, Theorem 3.10 scope, and raw Π^{Θ}-descent.

Thought for 7m 33s

According to a document from April 28, 2026, the strongest version of this whitepaper should make the following spine unmistakable:

$$\text{protected index} \to \text{Borcherds determinant} \to \text{GN denominator algebra} \to \text{physica}$$

The draft already contains most of the right architecture: it explicitly says the half-index is only a K_0-package, distinguishes the OP/DT scalar square from the compact Hall problem, introduces Γ_X^{phys}, Π, B, and a cocycle-corrected descent, and frames positive elliptic degree as a genuine locality obstruction rather than a bug.

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Below is an exhaustive technical upgrade list: fatal fixes first, then structural rewrites, then new insertions, then section-by-section changes.

0. Highest-priority surgical fixes

0.1 Replace Lemma 5.1 completely

Current problem: Lemma 5.1 says "smooth projective fourfold," leaves a vague "Todd correction," and therefore weakens the most important physical dictionary in the paper. $K3 \times E$ is a Calabi–Yau threefold, and the OP variable $n = \chi(\mathcal{O}_Y)$ can be made exact. The draft currently states

$$v_X(I_Y) = (1, 0, 1-d) \otimes 1_E + (0, -\beta_h, -k) \otimes \omega_E$$

with $k = n - \text{Todd correction}.$ raeez.lorgat.automorphic-correc...

Replace it with:

> **Lemma. D6–D2–D0 Mukai–Gram dictionary.**
> Let $X = S \times E$, with S a K3 surface and E an elliptic curve. Let $Y \subset X$ be a one-dimensional subscheme with
>
> $$[Y] = (\beta, d) \in H_2(S, \mathbb{Z}) \oplus H_2(E, \mathbb{Z}), \qquad \chi(\mathcal{O}_Y) = n.$$
>
> Let I_Y be its ideal sheaf. With
>
> $$\mathrm{td}(S) = (1, 0, 1), \qquad \mathrm{td}(E) = 1,$$
>
> the Mukai vector decomposes as
>
> $$v_X(I_Y) = (1, 0, 1-d) \otimes 1_E + (0, -\beta, -n) \otimes \omega_E.$$
>
> Hence, setting
>
> $$P_Y = (1, 0, 1-d), \qquad Q_Y = (0, -\beta, -n),$$
>
> one obtains
>
> $$\Pi(Q_Y, P_Y) = \left(\frac{Q_Y^2}{2}, Q_Y \cdot P_Y, \frac{P_Y^2}{2} \right) = \left(\frac{\beta^2}{2}, n, d-1 \right).$$
>
> In particular, for $\beta = \beta_h$ with $\beta_h^2 = 2h - 2$,
>
> $$\Pi(Q_Y, P_Y) = (h-1, n, d-1).$$

This is a central theorem, not a side lemma. It proves the OP/Borcherds variables from physical Mukai data.

0.2 Correct the BE cohomology in Lemma 5.8

Current problem: Lemma 5.8 treats $H^1(BE;\mathbb{Z}/2)$ as $(\mathbb{Z}/2)^2$. That is false for connected $E^{\mathrm{top}} \simeq T^2$. The calculation shown belongs to $BE[2]$, not BE. The draft later correctly treats $E[2]$ as Klein-four, but the connected E-quotient and finite stabilizer quotient are being mixed. raeez.lorgat.automorphic-correc... raeez.lorgat.automorphic-correc...

Replace Lemma 5.8 with two lemmas.

Replacement A: connected E-descent

$$E^{\mathrm{top}} \simeq T^2, \qquad BE \simeq BT^2.$$

Therefore

$$H^*(BE;\mathbb{F}_2) = \mathbb{F}_2[u_1,u_2], \qquad |u_i| = 2.$$

So

$$H^1(BE;\mathbb{F}_2) = 0, \qquad H^2(BE;\mathbb{F}_2) = \mathbb{F}_2 u_1 \oplus \mathbb{F}_2 u_2.$$

Thus there is no $E_2^{1,1}$ obstruction from $H^1(BE)$. The connected equivariant obstruction sits in total degree two through

$$E_2^{2,0} = H^2(BE;\mathbb{F}_2) \otimes H^0(\mathfrak{M}_C;\mathbb{F}_2)$$

and

$$E_2^{0,2} = H^0(BE; H^2(\mathfrak{M}_C;\mathbb{F}_2)).$$

Replacement B: finite $E[N]$-stabilizer descent

For $N = 2$,

$$E[2] \cong (\mathbb{Z}/2)^2,$$

and

$$H^*(BE[2];\mathbb{F}_2) = \mathbb{F}_2[x_1,x_2], \qquad |x_i| = 1.$$

Therefore

$$H^2(BE[2];\mathbb{F}_2) = \mathbb{F}_2\langle x_1^2, x_1x_2, x_2^2\rangle.$$

This is where the rank-three obstruction group belongs.

0.3 Downgrade Lemma 5.9 from “vanishing” to “criterion”

Current problem: Lemma 5.9 says translation acts trivially on $\mathrm{Pic}(E)$, $H^*(E)$, and Ω^1_E, so the $E[2]$-gerbe residual vanishes. But a line bundle can be fixed as an underlying line and still carry a nontrivial equivariant linearization. The obstruction is precisely equivariant.

Replace the theorem with:

> **Lemma. Klein-four residual criterion.**
> Let
>
> $$\beta_C^{E,E[2]} \in H^2(BE[2];\mathbb{F}_2)$$
>
> be the residual small-orbit orientation-gerbe class. Let
>
> $$\chi_{o_C} : E[2] \to \{\pm 1\}$$
>
> be the character of the $E[2]$-linearization of the reduced orientation line o_C^{red}. If χ_{o_C} restricts trivially to each of the three nontrivial cyclic subgroups of $E[2]$, and if the associated degree-two class vanishes under the three cyclic restrictions, then
>
> $$\beta_C^{E,E[2]} = 0.$$
>
> Conversely, nontrivial $E[2]$-linearization may produce a nonzero residual class even when the underlying ordinary line is translation-invariant.

Then state the actual computation as an open rank-one check.

0.4 Remove the Bott-element explanation of $B(c,c) = 2\Pi(c)$

The identity

$$B(c,c) = 2\Pi(c)$$

is pure lattice polarization. It does not depend on Bott periodicity, orientation lines, or parity conventions. The current Bott-element paragraph risks making an elementary

lattice identity look contingent on K-theoretic sign choices.

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Replace the whole Remark 4.4 with:

> **Remark. Polarization is independent of orientation.**
> The identity $B(c,c) = 2\Pi(c)$ is the polarization identity for the quadratic Gram map Π. It is independent of orientation data, Bott-periodicity conventions, and the parity split of any protected state space. Orientation choices enter only later, when one tries to promote the determinant to a Pfaffian section.

Move all Bott/Pfaffian sign conventions into the orientation section.

0.5 Stop saying the OP minus sign is sourced by the reduced parity flip

The draft now correctly says constants are not orientation data, but elsewhere it still says the cosection-induced parity flip is the sign that makes

$$Z_{\mathrm{OP}}^{X} = -4096\Delta_5^{-2}$$

carry its overall minus. raeez.lorgat.automorphic-correc...

That is too strong. The OP minus is a scalar branch convention. The Hall orientation character is a monodromy character. A scalar sign cannot determine a reflection character.

Replace every such sentence with:

> The OP minus sign is a scalar convention of the reduced OP/DT branch. It may be compatible with a chosen fermionic grading convention, but it is not evidence for the Hall orientation character. The reflection character ϵ_o must be computed from monodromy of the orientation line around type-II walls.

This aligns with the stronger statement already present in Remark 6.5.

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1. Make the normal-ordered charge extension the central new mathematics

The draft has the right material:

$$\Gamma_X^{\mathrm{phys}} = \Lambda_S \oplus \Lambda_S,$$

$$\Pi(Q,P) = \left(Q^2/2, Q\cdot P, P^2/2\right),$$

$$B(c,c') = (Q\cdot Q', Q\cdot P' + Q'\cdot P, P\cdot P'),$$

and

$$(c,T)\star(c',T') = (c+c', T+T'+B(c,c')).$$

It also correctly states this is degree bookkeeping, not a chain-level strictification.

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But the strongest version should make this the paper's **main original theorem package**.

1.1 Rename the extension

Use one notation consistently:

$$\Gamma_X := \Gamma_X^{\mathrm{phys}} \oplus_B \Gamma_{\mathrm{gram}}.$$

Avoid switching among Γ_X, Γ_X, $b\Gamma_X$, and $\Gamma_X^{\mathrm{phys}} \oplus_B \Gamma_{\mathrm{ind}}$.

Recommended convention:

$$\Gamma_X^{\mathrm{phys}} = \Lambda_S \oplus \Lambda_S,$$

$$\Gamma_{\mathrm{gram}} = \mathbb{Z}^3,$$

$$\Gamma_X = \Gamma_X^{\mathrm{phys}} \oplus_B \Gamma_{\mathrm{gram}}.$$

1.2 Define the normal-ordered Gram map

Insert immediately after the $\star$-product:

$$\Pi : \Gamma_X \to \Gamma_{\mathrm{gram}}, \qquad \Pi(c,T) = \Pi(c) - T.$$

Then prove:

$$\Pi((c,T)\star(c',T')) = \Pi(c,T) + \Pi(c',T').$$

This makes the descent completely explicit. It is clearer than saying "Θ subtracts B."

1.3 Demote Θ to categorification

Use:

$$\Pi$$

for the rigorous lattice-level theorem, and use

$$\Theta$$

only for the chain-level/categorical operation realizing this normal-ordering inside the Hall/factorization category.

Suggested phrasing:

> At the lattice level, normal-ordering is the homomorphism $\Pi(c, T) = \Pi(c) - T$. At the Hall-chain level, Θ is the additional datum required to realize the same subtraction as coherent degree-shift isomorphisms on bracket targets.

This is already implicit in the current Θ-descent discussion.

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2. Add a no-go theorem for raw Π-descent

This is one of the most important possible upgrades. It proves your cocycle-corrected descent is not decorative.

Insert after Lemma 7.7 or before Theorem 10.18.

> **Theorem. Raw Gram descent cannot realize the full BKM real-root strings.**
> Let
>
> $$P = \bigoplus_{c \in \Gamma_X^{\mathrm{phys}}} P_c$$
>
> be a physical-charge-graded Lie superalgebra with
>
> $$[P_c, P_{c'}] \subset P_{c+c'}.$$

Suppose raw Π-pushforward were Γ_{gram}-graded. Then every nonzero bracket would require

$$B(c, c') = 0.$$

In the BKM algebra $\mathfrak{g}_{\Delta_5}$, for distinct real simple roots e_i, e_j, one has

$$[e_i, e_j] \neq 0, \qquad [e_i, [e_i, e_j]] \neq 0, \qquad (\operatorname{ad} e_i)^3 e_j = 0.$$

Let c_i, c_j be physical lifts. The first nonzero bracket forces

$$B(c_i, c_j) = 0.$$

The second nonzero bracket forces

$$B(c_i, c_i + c_j) = 0.$$

But

$$B(c_i, c_i + c_j) = B(c_i, c_i) + B(c_i, c_j) = 2\Pi(c_i) \neq 0.$$

Contradiction. Therefore the full BKM bracket cannot be obtained by raw Π-descent.

Then add:

$$\text{The BKM comparison must use normal-ordered/cocycle-corrected Gram descent.}$$

This makes your physical charge-lattice correction a theorem of necessity, not merely bookkeeping. The BKM real-root data and Cartan matrix are already in the paper.

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3. Insert the primitive-lift lemma for every Gram triple

This strengthens the physical charge-lattice section.

Lemma. Primitive lift of every Gram triple.
Suppose Λ_S contains two orthogonal hyperbolic planes $U_1 \oplus U_2$, with bases e_i, f_i, $e_i^2 = f_i^2 = 0$, $e_i \cdot f_i = 1$. For every

$$(n, l, m) \in \mathbb{Z}^3$$

set

$$Q = e_1 + n f_1, \qquad P = l f_1 + e_2 + m f_2.$$

Then

$$Q^2 = 2n, \qquad Q \cdot P = l, \qquad P^2 = 2m.$$

Hence

$$\Pi(Q, P) = (n, l, m).$$

Moreover the sublattice $\mathbb{Z}Q + \mathbb{Z}P$ is primitive because the coefficient matrix contains a 2×2 minor equal to 1.

This proves every formal Borcherds degree has an honest primitive physical charge-plane lift. The problem is not existence of charges; the problem is compatibility with addition and Hall brackets.

4. Reframe Theorem 3.10

The current Theorem 3.10 is the most over-ambitious part of the draft. It assumes orientation data and a Pfaffian local model, then concludes that the first-order protected operator/algebra $\mathfrak{D}_X$ exists. But orientation data alone does not construct a chiral protected module, a spectrum, a primitive bracket, or a compact factorization object. The theorem currently risks turning an open construction into a definitional assumption.

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Split it into four pieces.

4.1 Definition: Dirac–Igusa datum

Insert:

$$\mathfrak{D}_X^{\mathrm{Ig}} = \left(\mathcal{A}_X^{E_3}, \mathcal{F}_X^{\mathrm{hyb}}, \Gamma_X, \Pi, o, \epsilon_o, P_X, \Theta_{\mathrm{rec}}\right).$$

Then define each component:

1. $\mathcal{A}_X^{E_3}$: compact protected holomorphic factorization object on X.
2. $\mathcal{F}_X^{\mathrm{hyb}}$: hybrid $K3 \to E$ chiral/wrapped shadow.
3. Γ_X: normal-ordered charge extension.
4. Π: additive Gram-degree map.
5. o: orientation/Pfaffian line.
6. ϵ_o: orientation-monodromy character.
7. $P_X = H^0 \operatorname{Prim}^{\mathrm{prot}}(\mathcal{F}_X^{\mathrm{hyb}})$: primitive protected algebra.

8. Θ_{rec}: recognition comparison with $\mathfrak{g}_{\Delta_5}$.

4.2 Theorem: Pfaffian determinant conditional on recognition

If a Dirac–Igusa datum exists and

$$\mathrm{sdim}(\Pi_* P_X)_{(n,l,m)} = f(nm, l),$$

then

$$\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X) = \Delta_5.$$

This is rigorous and honest.

4.3 Theorem: scalar square

If

$$\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X) = \Delta_5,$$

then the OP scalar branch is

$$Z^X_{\mathrm{OP/DT}} = -4096\ \mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X)^{-2}.$$

This follows from the scalar theorem already in the paper. raeez.lorgat.automorphic-correc...

4.4 Theorem: orientation character

If ϵ_o exists as a group character and

$$\epsilon_o(s_{\delta_i}) = -1, \qquad i = 1, 2, 3,$$

then

$$\epsilon_o = \nu_{\Delta_5}$$

on $W^{(2)}(\Lambda^{2,1}_{II})$.

This is a clean group-theoretic theorem.

5. Simplify the orientation-character section

The current proof around Lemma 3.11 is too long and too risky. The three-wall cocycle discussion creates technical liabilities. The core theorem is much simpler.

5.1 Keep this

The Maass character has value -1 on the three type-II simple reflections and $+1$ on the chamber-permuting S_3 reflections. This is already stated in the Weyl/multiplier section.

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5.2 Replace the long proof with

> The group $W^{(2)}(\Lambda_{II}^{2,1})$ is generated by the three simple reflections $s_{\delta_1}, s_{\delta_2}, s_{\delta_3}$. Since the Coxeter exponents between distinct simple reflections are infinite, its abelianization is $(\mathbb{Z}/2)^3$. Hence a $\{\pm 1\}$-character is determined by its values on these three generators. Therefore, if the geometric orientation monodromy satisfies $\epsilon_o(s_{\delta_i}) = -1$ for $i = 1, 2, 3$, then ϵ_o equals the determinant character on $W^{(2)}$. Since ν_{Δ_5} has the same values on the same generators, $\epsilon_o = \nu_{\Delta_5}$.

5.3 State the actual open calculation

$$\epsilon_o(s_{\delta_i}) = -1$$

must be computed from determinant-line monodromy of the compact reduced Hall sector. That is the geometric theorem, not the Maass-side computation.

6. Make the hybrid wrapped factorization object concrete

The current paper correctly states that positive elliptic degree projects onto all of E, so ordinary Ran(E)-support locality cannot see the full s-direction.

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Level it up by defining an actual target category.

6.1 Define $\mathrm{Ran}^{\mathrm{hyb}}(E)$

Objects:

$$(F, b),$$

where

$$F \subset E$$

is a finite subset and

$$b \in \mathbb{Z}_{\geq 0}$$

is total elliptic wrapping degree.

The $b = 0$ stratum is ordinary Ran locality.

The $b > 0$ strata are global wrapped sectors.

6.2 Define the hybrid factorization object

$$\mathcal{F}_X^{\mathrm{hyb}}(F, b).$$

Require:

$$\mathcal{F}_X^{\mathrm{hyb}}(F, 0) = \mathcal{F}_X^{\mathrm{pf}}(F),$$

the projection-finite local sector.

For $b > 0$,

$$\mathcal{F}_X^{\mathrm{hyb}}(F, b)$$

is a module/sector for wrapped elliptic-degree objects.

6.3 Define structure maps

For disjoint finite supports:

$$\mathcal{F}_X^{\mathrm{hyb}}(F_1, b_1) \otimes \cdots \otimes \mathcal{F}_X^{\mathrm{hyb}}(F_k, b_k) \to \mathcal{F}_X^{\mathrm{hyb}}\left(\bigcup_i F_i,\ b_1 + \cdots + b_k + \mathrm{interaction}\right).$$

The interaction is recorded upstairs in Γ_X^{phys} and then normal-ordered through Π.

6.4 Tie this to Open Problem 10.15

The draft already asks for a hybrid object with projection-finite, wrapped, and mixed correspondences. raeez.lorgat.automorphic-correc... Turn that from prose into a definition.

7. Clarify "holomorphic E_3" in three layers

The phrase "holomorphic E_3-factorization algebra" is physically suggestive but mathematically underdefined.

Use a layered definition.

7.1 First layer: holomorphic factorization

$$\mathcal{A}_X \in \mathrm{Fact}^{\mathrm{hol}}(X; \mathrm{Ch}_{\mathbb{C}}).$$

This is the primary object.

7.2 Second layer: local E_3-shadow

For $x \in X$, a holomorphic polydisc $D_x \simeq \mathbb{C}^3$, compact-support conditions, and a formality/framing datum produce

$$\mathcal{A}^{\mathrm{loc}}_{X,x} \in E_3\text{-}\mathrm{Alg}.$$

7.3 Third layer: $K3 \to E$ specialization

$$\mathrm{Sp}^{\mathrm{ch}}_{K3,E}(\mathcal{A}_X)$$

is not a slogan; it must be a functor/correspondence with:

1. source category,
2. target category,
3. support conditions,
4. compatibility with charge grading,
5. compatibility with orientation,
6. compatibility with wrapped sectors.

The current document already says $\Phi_3^{FA}(X)$ is a placeholder and no external formalism is used as an existence theorem unless it supplies chain-level observables, factorization maps, QME/anomaly control, and compact charge completion. Keep this warning.

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8. Recast the compact realization datum

Current Definition 10.29/10.30 is too large and partly tautological: it assumes a primitive comparison certificate so strong that the conclusion is essentially forced. That is acceptable only if called a certificate, not a theorem producing structure.

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8.1 Rename it

Change:

> Compact Igusa realization datum

to:

> Dirac–Igusa realization certificate.

8.2 Split it into levels

Use five levels:

$$\mathfrak{C}_X = (L_X, H_X, O_X, D_X, R_X).$$

Where:

1. L_X: local protected observable/factorization source.
2. H_X: hybrid $K3 \to E$ Hall/chiral shadow.
3. O_X: orientation/Pfaffian and equivariant descent.
4. D_X: normal-ordered Gram descent.
5. R_X: primitive BKM recognition certificate.

8.3 State what each level proves

- L_X gives locality.
- H_X gives wrapped-sector collision maps.
- O_X gives Pfaffian signs.
- D_X gives additive Gram grading.
- R_X gives $\mathfrak{g}_{\Delta_5}$.

This prevents the reader from thinking one piece does all the work.

9. Fix sectorial Hall truncation to use the physical lattice

Current sectorial Hall truncation sometimes assumes a numerical Gram-index lattice Γ_{ind}. That is incompatible with the physical charge-lattice correction. The draft elsewhere knows this, but the truncation section must be made consistent.

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Replace:

$$\gamma \in \Gamma_{\mathrm{ind}}$$

by either:

$$c \in \Gamma_X^{\mathrm{phys}}$$

or

$$c \in \Gamma_X.$$

Then define the height function on the additive lattice:

$$h_S(c) = \mathrm{Re}(e^{-i\vartheta} Z_\sigma(c)).$$

The Gram-index height can appear only after normal-ordered pushforward.

10. Add the normal-ordered primitive recognition theorem

Replace the current recognition theorem with:

> **Theorem. Normal-ordered primitive recognition.**
> Suppose a compact Dirac–Igusa realization supplies a Γ_X-graded protected primitive Lie superalgebra
>
> $$P_X = H^0 \operatorname{Prim}^{\mathrm{prot}}(\mathcal{F}_X^{\mathrm{hyb}})$$
>
> with
>
> $$[P_c, P_{c'}] \subset P_{c \star c'}.$$
>
> Then
>
> $$\Pi_* P_X$$
>
> is a Γ_{gram}-graded Lie superalgebra. If the BKM recognition certificate holds for $\Pi_* P_X$, then
>
> $$\Pi_* P_X \cong \mathfrak{g}_{\Delta_5}.$$

This avoids raw Π-grading errors and makes the role of B precise.

11. Clean up the “Dirac” language

The Dirac inspiration is strong, but it should be consistently mathematical.

11.1 Preferred formulation

Use:

$$\text{The square root is first-order structure, not analytic branch choice.}$$

Then:

$$\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X) = \Delta_5$$

and

$$Z^X_{\mathrm{OP/DT}} = -4096\ \mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X)^{-2}.$$

11.2 Avoid saying

- "Δ_5 is the square root of OP/DT" without qualification.
- "orientation existence realizes $\mathfrak{D}_X$" unless $\mathfrak{D}_X$ is separately constructed.
- "the OP sign pins the Berezinian convention."

11.3 Use instead

> The OP/DT scalar branch is compatible with a first-order Pfaffian object, but does not construct it. The first-order object requires a compact protected algebra, orientation, normal-ordered Gram descent, and primitive recognition.

12. Add a precise "theorem-status ledger" before the abstract or after it

The paper already has a claim-strength table. Keep it, but sharpen it.

Use columns:

Object	Status	Source	What it proves	What it does not prove
K3 elliptic genus	imported	K3 sigma	protected	compact $K3 \times E$
$Z_{K3} = 2\phi_{0,1}$	theorem	model/EZ	index	Hilbert space

Object	Status	Source	What it proves	What it does not prove
K_0-determinant $D_X = \Delta_5$	proved from imported Borcherds product	this paper + GN	normalized product	operator/Pfaffian
BKM algebra $\mathfrak{g}_{\Delta_5}$	imported theorem	GN	target bracket	compact Hall bracket
OP scalar square	imported theorem	OP/Oberdieck	scalar reduced series	orientation character
$\Gamma_X^{\mathrm{phys}}, B, \Gamma_X$	proved here	lattice algebra	correct descent lattice	compact realization
Θ-descent	conditional	compact Hall chain data	Gram-graded bracket	construction of Θ
$\mathfrak{D}_X$	open	Dirac–Igusa datum	Pfaffian target	existence

This makes the paper referee-proof.

13. Abstract rewrite

Current abstract is strong but still too long and theorem-heavy. Suggested replacement:

> For $X = K3 \times E$, the protected K3 index satisfies $Z_{K3} = 2\phi_{0,1}$. Replacing it by the Borcherds half $\phi_{0,1}$ gives a K_0-level virtual determinant, not a microscopic Hilbert space. We prove that the theta-normalized determinant is
>
> $$D_X(Z) = 64 q^{1/2} r^{1/2} s^{1/2} \prod_{\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)} = \Delta_5(Z).$$
>
> The same coefficients are the signed root supermultiplicities of the Gritsenko–Nikulin denominator algebra $\mathfrak{g}_{\Delta_5}$, with

$$\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z).$$

The reduced OP/DT scalar branch gives

$$Z^X_{\mathrm{OP/DT}} = -4096\Delta_5^{-2},$$

but this scalar square forgets the Maass reflection character.

The central structural problem is physical: a compact Hall/factorization realization is not graded by $\mathbb{Z}^3$ directly. The additive physical lattice is

$$\Gamma_X^{\mathrm{phys}} = H(K3,\mathbb{Z}) \oplus H(K3,\mathbb{Z}),$$

and the Igusa variables are the quadratic Gram shadow

$$\Pi(Q,P) = \left(Q^2/2, Q\cdot P, P^2/2\right).$$

We isolate the exact additivity defect

$$B(c,c') = (Q\cdot Q', Q\cdot P' + Q'\cdot P, P\cdot P')$$

and construct the normal-ordered extension $\Gamma_X = \Gamma_X^{\mathrm{phys}} \oplus_B \Gamma_{\mathrm{gram}}$ on which the corrected Gram map is additive. Raw Π-descent cannot realize the full BKM real-root strings; cocycle-corrected descent is forced.

The square root is therefore meant in Dirac's sense: the open problem is to construct a first-order, orientation-sensitive protected algebra whose Pfaffian is Δ_5, and whose orientation-forgetting scalar square is the OP/DT branch.

14. Section 1 recommendations

14.1 Replace "local object comes first" with a theorem-program diagram

Use:

$$
\begin{aligned}
&\text{worldsheet protected index}\\
&\Downarrow\\
&K_0\text{-Borcherds determinant}\\
&\Downarrow\\
&\text{BKM denominator target}\\
&\Downarrow\\
&\text{physical lattice } \Gamma_X^{\mathrm{phys}} \text{ and normal-ordered Gram descent}\\
&\Downarrow\\
&\text{compact Dirac–Igusa realization problem.}
\end{aligned}
$$

14.2 Add “What is new here?”

Separate:

- new in manuscript:
  - exact B-cocycle formalism,
  - normal-ordered charge extension,
  - no-go theorem for raw Π-descent,
  - hybrid locality formulation,
  - Dirac–Igusa certificate.
- imported:
  - K3 elliptic genus,
  - Borcherds/GN product,
  - GN BKM denominator,
  - OP/Oberdieck scalar square.
- open:
  - compact E_3 source,
  - hybrid wrapped Hall category,
  - orientation monodromy,
  - primitive BKM recognition.

15. Section 2 recommendations: physical question

The section is conceptually good. It says charged BPS states are sectors, not local fields, and locality belongs to protected operations. Keep this.

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15.1 Add a paragraph distinguishing spacetime locality from support-locality

Suggested text:

> The obstruction below is not a failure of spacetime locality in the six-dimensional/topological-holomorphic theory. It is a failure of support-locality after projection to E. A positive elliptic-degree brane is local as a brane in X, but it is not local as a point insertion on E.

15.2 Replace $F_X \in \mathrm{Fact}(\mathrm{Ran}(E))$

Use:

$$F_X^{\mathrm{pf}} \in \mathrm{Fact}(\mathrm{Ran}(E))$$

for projection-finite sectors, and

$$F_X^{\mathrm{hyb}} \in \mathrm{Fact}^{\mathrm{hyb}}(E)$$

for the full object.

15.3 Avoid saying the determinant gives "the one-particle index such a realization would have"

Use:

> The determinant gives the signed Gram-index shadow that the primitive normal-ordered pushforward must reproduce.

16. Section 3 recommendations: one-particle index and determinant

16.1 Rename "one-particle object"

Use:

$$V_{\mathrm{Borch}}^{K_0}$$

or

$$V_{\mathrm{virt}}$$

instead of “one-particle object,” unless every occurrence is immediately qualified.

16.2 Keep the split

The split

$$V = V_{\mathrm{bulk}} \oplus V_{\mathrm{cusp}}$$

is excellent. It prevents misreading $m = 0$ as physical one-particle states.

16.3 Replace $Z^{\mathrm{vir}}_{\mathrm{BPS}}$

Use:

$$Z^{\mathrm{vir}}_{\mathrm{Borch}}$$

throughout.

“BPS” invites the exact confusion the paper is trying to avoid.

16.4 Recast Proposition 3.9

Make Proposition 3.9 a **definition of target**, not a theorem.

Suggested name:

> Definition. Dirac/Pfaffian target problem.

16.5 Split Theorem 3.10

As above: do not claim orientation existence realizes $\mathfrak{D}_X$. Orientation is necessary, not sufficient.

17. Section 4 recommendations: normalized cusp form

17.1 Keep normalization table

The normalization table is useful. It fixes:

$$[q^{1/2} r^{1/2} s^{1/2}]\Delta_5 = 64, \qquad D_5 = 64^{-1}\Delta_5.$$

17.2 Delete Bott remark

Move all orientation/Pfaffian sign material elsewhere.

17.3 Add “automorphic line” emphasis

Every occurrence of Δ_5 should be understood as a section:

$$\Delta_5 \in H^0(\mathbb{H}_2, \mathcal{L}^5 \otimes \nu_{\Delta_5}).$$

The scalar square lies in:

$$\Delta_5^2 \in H^0(\mathbb{H}_2, \mathcal{L}^{10}).$$

This strengthens the Dirac/Pfaffian thesis.

18. Section 5 recommendations: protected-index dictionary

18.1 Replace Lemma 5.1 as above

This is the most urgent local fix.

18.2 Move physical charge lattice earlier

Put the dyonic Mukai lattice before OP/DT scalar integration. The reader should see:

$$v_X(I_Y) = P \otimes 1_E + Q \otimes \omega_E$$

before the scalar quotient.

18.3 Rephrase rank-one orientation claims

Any claim like “orientation existence constructed” should be weakened unless you explicitly prove the reduced quotient orientation line, its descent, and its equivariant linearization.

Use:

> Rank-one orientation is expected/reduced to known Pin^c input plus a reduced quotient descent check.

18.4 Split orientation problem into four subproblems

Use:

$$O_{\mathrm{CY3}}$$

unreduced CY3 orientation,

$$O_{\mathrm{red}}$$

cosection-reduced orientation,

$$O_E$$

free E-quotient descent,

$$O_{\mathrm{orb}}$$

finite stabilizer/small-orbit descent,

$$O_W$$

Weyl-equivariant orientation monodromy.

Then:

$$O_{\mathrm{CY3}} \not\Rightarrow O_{\mathrm{red}} \not\Rightarrow O_E \not\Rightarrow O_W.$$

18.5 Fix equivariant descent as above

Do not mix BE and $BE[2]$.

19. Section 6 recommendations: scalar square

This section is mostly good.

19.1 Replace “analytic root” language

Current text says $D_X = \Delta_5$ is one analytic root and the opposite-sign root is also a root. raeez.lorgat.automorphic-correc...

Use:

> The Borcherds product supplies a normalized automorphic section $D_X = \Delta_5$. The scalar OP/DT branch sees only its square. The opposite sign is not a second physical solution unless supplied by an orientation convention; it is the same scalar square with the Pfaffian orientation forgotten.

19.2 Put OP sign caveat in theorem statement

Add to Theorem 6.2:

> The leading minus is part of the OP scalar normalization and is not a Hall orientation character.

19.3 Add a one-line physics reading

$$\mathrm{OP/DT\ proves\ the\ second\text{-}order\ scalar\ amplitude,\ not\ the\ first\text{-}order\ protected\ alg}$$

20. Section 7 recommendations: duality principle and physical lattice

This is one of the strongest parts.

20.1 Distinguish algebraic Mukai sector from full $N = 4$ charge lattice

The draft says $\Gamma_X^{\mathrm{phys}} = \Lambda_S \oplus \Lambda_S$ is the physical microscopic charge lattice. That is too broad physically. It is the **algebraic even Mukai sector** relevant to the D6–D2–D0/OP branch.

Replace with:

> In the algebraic even D-brane sector relevant to the D6–D2–D0/OP branch, the additive physical charge lattice is $H(S, \mathbb{Z}) \oplus H(S, \mathbb{Z})$. In the full $N = 4$ compactification this embeds into a larger electric-magnetic charge lattice. The genus-two Igusa variables are the Gram invariants of a rank-two charge plane.

20.2 Insert primitive-lift lemma

As above.

20.3 Insert normal-ordered Gram map

As above.

20.4 Add flag formula

> For elementary charges $c_1, \ldots, c_k$,

$$(c_1,0)\star\cdots\star(c_k,0)=\left(\sum_i c_i,\sum_{i<j}B(c_i,c_j)\right).$$

Hence

$$\Pi((c_1,0)\star\cdots\star(c_k,0))=\sum_i \Pi(c_i).$$

This gives a physical interpretation: the BKM/Fock degree is the normal-ordered Gram degree of an extension flag.

21. Section 8 recommendations: exterior-square model

21.1 Add a warning on PSp_4 vs full physical duality

Use:

> The group $PSp_4(\mathbb{Z})$ here is the automorphic group of the genus-two variable and the invariant Gram plane. It is not asserted to be the full microscopic U-duality group of the compactification.

21.2 Check all matrix conventions

Add an appendix table matching:

$$(q,r,s),\quad (z_1,z_2,z_3),\quad (f_2,f_3,f_{-2}),\quad (n,l,m),\quad \alpha(n,l,m).$$

22. Section 9 recommendations: Weyl chamber

22.1 Keep the active support lemma

The lemma proving

$$\alpha(\gamma)=n\delta_1+m\delta_2+(n+m-l)\delta_3$$

is valuable.

22.2 Add “active support is not additive”

This fact appears later. Put it here too.

Example:

$$(0,-1,0) \in \Gamma_{\mathrm{act}}, \qquad 2(0,-1,0) = (0,-2,0) \notin \Gamma_{\mathrm{act}}.$$

This supports why Hall truncation cannot be active-support finite.

22.3 Separate real-root divisor signs from orientation signs

The automorphic sign

$$\nu_{\Delta_5}(s_\delta)$$

is theorem-level. The Hall sign

$$\epsilon_o(s_\delta)$$

is open. Never conflate them.

23. Section 10 recommendations: denominator algebra and compact realization

23.1 Rename A_{den}

Use:

$$\mathfrak{g}_{\Delta_5}$$

for the denominator algebra.

Use:

$$\mathrm{Cur}_E(\mathfrak{g}_{\Delta_5})$$

for the current object.

Avoid names suggesting it already came from $K3 \times E$.

23.2 Formal current envelope caveat

Keep emphasizing:

$$\mathrm{Cur}_E(\mathfrak{g}_{\Delta_5})$$

is a formal target construction. It does not use compact geometry except the chosen curve E.

23.3 Replace Γ_{ind}-graded primitive recognition

Use normal-ordered Π-pushforward.

23.4 Add no-go theorem

As above.

23.5 Reduce tautology in primitive recognition theorem

Instead of assuming:

- simple representatives,
- all defining relations,
- no extra relations,
- PBW compatibility,
- full parity dimensions,

and then calling it a theorem, call it:

> Recognition certificate.

The "theorem" should simply say the certificate is sufficient.

23.6 Add "minimal recognition" vs "full recognition"

Minimal recognition:

$$\operatorname{sdim} P_\gamma = f(nm, l)$$

is necessary.

Full recognition:

$$P \cong \mathfrak{g}_{\Delta_5}$$

requires bracket, parity, relations, PBW, radical quotient.

23.7 Explicitly distinguish three brackets

1. Hall bracket upstairs:

$$[\,,\,]_{\mathrm{Hall}} : P_c \otimes P_{c'} \to P_{c+c'}.$$

2. Normal-ordered descended bracket:

$$[\,,\,]_\Theta : (\Pi_* P)_\gamma \otimes (\Pi_* P)_{\gamma'} \to (\Pi_* P)_{\gamma+\gamma'}.$$

3. BKM bracket:

$$[\,,\,]_{\mathrm{BKM}} : \mathfrak{g}_\alpha \otimes \mathfrak{g}_\beta \to \mathfrak{g}_{\alpha+\beta}.$$

The comparison is between 2 and 3, not between 1 and 3 directly.

24. Section 11 recommendations: synthesis

The synthesis should become the paper's final theorem package.

24.1 Replace "Coefficient dictionary and synthesis theorem"

Use:

> Main theorem package and open Dirac–Igusa realization problem.

24.2 Include all four theorem tiers

1. K_0-determinant:

$$D_X = \Delta_5.$$

2. GN denominator:

$$\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z).$$

3. OP scalar:

$$Z^X_{\mathrm{OP/DT}} = -4096\Delta_5^{-2}.$$

4. Normal-ordered realization target:

$$H^0\Pi_* \operatorname{Prim}^{\mathrm{prot}}(\mathcal{F}_X^{\mathrm{hyb}}) \overset{?}{\cong} \mathfrak{g}_{\Delta_5}.$$

24.3 Add exact "what would advance knowledge" paragraph

> The new mathematical object demanded by the physics is a normal-ordered, orientation-sensitive, hybrid local/global Hall factorization algebra on $K3 \times E$. Its primitive bracket should be physically graded by Γ_X^{phys}, normal-ordered through Γ_X, and Gram-descended to the Gritsenko–Nikulin BKM algebra.

25. Part II recommendations: eight rows

The row-certificate material is useful but risks overwhelming the $N = 1$ paper.

25.1 Make Part II explicitly independent

Add:

> Part II is not used in the proof of the $N = 1$ theorem. It is a certificate ledger for future boundary-row physics.

25.2 Put every row in a fixed certificate table

For each row:

$$(F_j, \Gamma_j, \text{cover}, \nu_j, \phi_j, L_j, \rho_j, \text{product theorem}, \text{denominator theorem}, \text{scalar theorem},$$

25.3 Avoid "physical status" unless a host exists

Use:

- automorphic product,
- denominator theorem,
- CHL scalar square,
- reduced CY/DT scalar host,
- compact Hall/chiral host.

Do not call rows physical unless the relevant certificate level is filled.

25.4 $F_7 = \nabla_{3/2}$

Always state:

$$F_7$$

lives on the μ_4-multiplier cover.

State exactly which square is single-valued and on what quotient.

26. Appendix A recommendations

The spin L-factor appendix is mathematically interesting but not structurally necessary.

26.1 Move it to a companion note

It is not used to construct:

$$\Gamma_X, \quad \Theta, \quad \mathfrak{D}_X, \quad \mathcal{F}_X^{\mathrm{hyb}}, \quad \epsilon_o, \quad \mathfrak{g}_{\Delta_5}.$$

26.2 If kept, add disclaimer

> This appendix is independent of the Dirac–Igusa construction. It records a scalar-square arithmetic normalization only.

27. New theorem package to insert

Add a section titled:

> The normal-ordered charge theorem.

It should include:

Theorem 1: physical charge lattice

$$\Gamma_X^{\mathrm{phys}} = H(S,\mathbb{Z}) \oplus H(S,\mathbb{Z}).$$

Theorem 2: Gram projection

$$\Pi(Q,P) = (Q^2/2, Q\cdot P, P^2/2).$$

Theorem 3: D6–D2–D0 dictionary

$$\Pi(Q_Y,P_Y) = (h-1, n, d-1).$$

Theorem 4: defect cocycle

$$\Pi(c+c') = \Pi(c) + \Pi(c') + B(c,c').$$

Theorem 5: normal-ordered extension

$$\Gamma_X = \Gamma_X^{\mathrm{phys}} \oplus_B \Gamma_{\mathrm{gram}}.$$

Theorem 6: additive normal-ordered map

$$\Pi(c,T) = \Pi(c) - T.$$

Theorem 7: raw descent no-go

Raw Π-descent cannot realize full $\mathfrak{g}_{\Delta_5}$.

Theorem 8: corrected recognition target

$$H^0\Pi_* \operatorname{Prim}^{\mathrm{prot}}(\mathcal{F}_X^{\mathrm{hyb}}) \overset{?}{\cong} \mathfrak{g}_{\Delta_5}.$$

This is the heart of the strongest paper.

28. Notation fixes

Use one name for each object.

Current ambiguity	Recommended
Γ_{ind}, Γ_{gram}	Use Γ_{gram}, define $\Gamma_{\mathrm{ind}} := \Gamma_{\mathrm{gram}}$ once only
Γ_X, $b\Gamma_X$, Γ_X	Use Γ_X
$Z^{\mathrm{vir}}_{\mathrm{BPS}}$	Use $Z^{\mathrm{vir}}_{\mathrm{Borch}}$
A_{den}	Use $\mathfrak{g}_{\Delta_5}$
D_X vs D_5	$D_X = \Delta_5$, $D_5 = 64^{-1}\Delta_5$
F_X	Use F_X^{pf} or F_X^{hyb}
Π_*^{Θ}	Define via Π_* first; Π_*^{Θ} only chain-level

29. Proof-hygiene fixes

29.1 Avoid proof sketches where theorem is strong

If a result says “vanishes,” give a proof. If proof is incomplete, call it a criterion or conjecture.

29.2 Separate algebraic proof from geometric assumption

For example:

- algebraic: B is a cocycle.
- geometric: Hall bracket realizes B-shift by Θ.

Do not put both under one theorem unless assumptions are explicit.

29.3 Do not use “standard” for nonstandard composites

"Standard equivariant Serre spectral sequence" is fine.

"Standard equivariant derived setting of reduced $K3 \times E$ Hall stack with cosection and orientation descent" is not standard.

29.4 Reduce citation stacking

Several paragraphs list many orientation references. Instead:

- one paragraph says what exists in known CY3/CY4 literature,
- next paragraph says what is not covered here.

29.5 Check dimensional language

Replace every "fourfold" reference to $K3 \times E$ with "threefold."

30. Physics-strength upgrades

30.1 Add a "Dirac principle" section

Suggested statement:

> A scalar square is not square-rooted by choosing a branch. It is square-rooted by finding a first-order object with extra algebraic degrees of freedom. In this paper those degrees of freedom are orientation, primitive bracket, and normal-ordered charge descent.

30.2 Add "unitarity before index, not after"

The draft already says this. Make it a boxed principle:

$$\text{unitarity proves localization; the localized object is cohomological and virtual.}$$

30.3 Add "states vs observables"

> BPS states are superselection sectors. The local/factorization object is the algebra of protected operations on sectors.

30.4 Add "wrapped sectors are line defects over E"

A positive elliptic-degree brane should be physically compared to a wrapped/global operator, not a point insertion on E.

30.5 Add “wall crossing”

State that the compact Hall category must be stability-chamber dependent, while Δ_5 is an automorphic object whose chamber expansion is selected by the cusp.

31. Mathematics-strength upgrades

31.1 Prove the raw descent no-go theorem

This is probably the most publishable new lemma beyond the cocycle itself.

31.2 Prove primitive lift/surjectivity of Π

Easy, useful, clarifies physical geometry.

31.3 Formalize Γ_X-graded Lie algebras

Define the category:

$$\mathrm{LieSAlg}_{\Gamma_X}$$

and the pushforward functor:

$$\Pi_* : \mathrm{LieSAlg}_{\Gamma_X} \to \mathrm{LieSAlg}_{\Gamma_{\mathrm{gram}}}.$$

31.4 Turn Θ-descent into a theorem about functors

At chain level:

$$\Theta_{c,c'} : P_{c+c'}[\Pi(c+c')] \to P_{c+c'}[\Pi(c)+\Pi(c')]$$

with coherence:

$$\Theta_{c+c',c''} \circ \Theta_{c,c'} = \Theta_{c,c'+c''} \circ \Theta_{c',c''}.$$

The equality is equivalent to $\delta B = 0$.

31.5 Define “normal-ordered PBW”

The PBW theorem in the recognition certificate should be compatible with Π, not raw Π.

32. Exact replacement for the compact recognition statement

Replace the current recognition theorem with this:

> **Theorem. Dirac–Igusa recognition certificate.**
> Suppose $X = K3 \times E$ is equipped with a compact hybrid protected factorization datum producing a Γ_X-graded primitive Lie superalgebra
>
> $$P_X.$$
>
> Suppose:
>
> 1. "$\Pi_* P_X$ has Cartan $\Lambda^{2,1}_{II} \otimes \mathbb{C}$;"
> 2. "its real simple primitive representatives have degrees $\delta_1, \delta_2, \delta_3$;"
> 3. "its imaginary simple primitive fibres have the GN parity and multiplicity data;"
> 4. "its bracket satisfies the Borcherds–Kac relations;"
> 5. "the completed Hopf pairing has the same radical quotient;"
> 6. "there are no extra completed relations;"
> 7. "completed PBW holds;"
> 8. "full parity dimensions match the GN algebra."
>
> Then
>
> $$\Pi_* P_X \cong \mathfrak{g}_{\Delta_5}.$$
>
> Consequently,
>
> $$\mathrm{den}(\Pi_* P_X) = 64^{-1} \Delta_5(2Z).$$

Then add:

> This theorem is a certificate theorem. Its assumptions are the compact realization problem.

33. Bibliography and source-status fixes

33.1 Distinguish sources by function

Create bibliography categories:

- Jacobi/K3 elliptic genus.
- Borcherds/Gritsenko–Nikulin.
- OP/Oberdieck.
- CoHA/Hall.
- PTVV/BBDJS/orientation.

- Factorization/chiral.
- CHL/boundary rows.

33.2 Avoid using sources for claims they do not prove

Example: orientation existence for unreduced CY3 stacks does not prove reduced E-quotient Weyl-equivariant orientation.

Say exactly:

> Existing orientation technology supplies the language and partial inputs; the reduced E-quotient Weyl-equivariant compact Hall orientation is not in the literature.

33.3 Add a "source role" note for each imported theorem

This prevents overclaim.

34. Title recommendations

Current title is good. Stronger possible titles:

1. **The Igusa Square Root as a Dirac Problem**
2. **Normal-Ordered Gram Descent and the Igusa Square Root**
3. **The Igusa Square Root: Borcherds Determinants, Physical Charge Lattices, and the $K3 \times E$ Dirac Problem**
4. **A Dirac–Borcherds Program for $K3 \times E$**

My recommendation:

> **The Igusa Square Root**
> **Borcherds determinants, normal-ordered charge descent, and the $K3 \times E$ Dirac realization problem**

35. Final "strongest possible" structure

Use this table of contents.

Part I. The automorphic square root

1. Claim status and Dirac principle
2. K3 protected index and K_0-half

3. Borcherds determinant $D_X = \Delta_5$
4. GN denominator algebra $\mathfrak{g}_{\Delta_5}$
5. OP/DT scalar square

Part II. Physical charge and normal ordering

6. Dyonic Mukai lattice
7. D6–D2–D0 Mukai–Gram dictionary
8. Gram defect cocycle B
9. Normal-ordered extension Γ_X
10. No-go theorem for raw Π-descent
11. Corrected primitive recognition target

Part III. Compact Dirac–Igusa realization problem

12. Local protected observables on X
13. $K3 \to E$ specialization
14. Projection-finite and wrapped locality
15. Hybrid wrapped factorization
16. Orientation and Pfaffian monodromy
17. Dirac–Igusa recognition certificate

Part IV. Boundary-row certificates

18. Eight diagonal-divisor rows
19. Product vs denominator vs scalar vs Hall host
20. Future row certificates

Appendices:

A. Normalizations and Fourier conventions
B. Lattice and Weyl chamber tables
C. Boundary-row ledger
D. Optional spin L-factor note, or remove

36. One-page executive rewrite to insert near the beginning

Use something close to:

The square root in this paper is meant in Dirac's sense. The scalar reduced OP/DT theory sees Δ_5^{-2}, but the first-order protected object should see Δ_5^{-1} with its Maass character. Such a first-order object cannot be recovered from the scalar square. It requires an orientation/Pfaffian line, a primitive bracket, and a local-to-global Hall/factorization structure.

The additive physical charge lattice is not $\mathbb{Z}^3$. It is

$$\Gamma_X^{\mathrm{phys}} = H(K3,\mathbb{Z}) \oplus H(K3,\mathbb{Z}).$$

The Igusa degrees are the quadratic Gram shadow

$$\Pi(Q,P) = \left(Q^2/2, Q\cdot P, P^2/2\right).$$

The failure of Π to be additive is the symmetric bilinear cocycle B. Therefore the correct compact comparison is not raw Π-pushforward. It is normal-ordered descent through

$$\Gamma_X = \Gamma_X^{\mathrm{phys}} \oplus_B \Gamma_{\mathrm{gram}}, \qquad \Pi(c,T) = \Pi(c) - T.$$

Raw Π-descent cannot realize the full real-root strings of $\mathfrak{g}_{\Delta_5}$. The cocycle correction is forced.

The main compact open problem is therefore to construct a hybrid local/global protected Hall factorization object on $K3 \times E$, with orientation and normal-ordered Gram descent, whose primitive algebra is the Gritsenko–Nikulin denominator algebra.

37. Final priority checklist

Must fix before sharing with serious referees

1. Correct Lemma 5.1.
2. Correct BE vs $BE[2]$.
3. Downgrade small-orbit gerbe vanishing to criterion.
4. Remove Bott explanation of $B(c,c) = 2\Pi(c)$.
5. Remove claim that OP minus is sourced by orientation/parity.
6. Split Theorem 3.10 into conditional pieces.
7. Make Γ_X, Π primary.
8. Add raw Π-descent no-go theorem.

9. Change Hall truncation from Γ_{ind} to Γ_X^{phys} or Γ_X.
10. Distinguish algebraic Mukai sector from full physical charge lattice.

Should fix for maximum strength

11. Define $\mathrm{Ran}^{\mathrm{hyb}}(E)$.
12. Formalize hybrid wrapped factorization.
13. Define holomorphic factorization first, E_3-shadow second.
14. Rename compact realization theorem as certificate.
15. Add primitive-lift lemma.
16. Add flag normal-ordering formula.
17. Simplify orientation-character proof.
18. Move spin L-appendix out or mark independent.
19. Add theorem-status ledger.
20. Add “what is new here” section.

Style/notation cleanup

21. Use Γ_{gram} consistently.
22. Use Γ_X consistently.
23. Rename $Z_{\mathrm{BPS}}^{\mathrm{vir}}$ to $Z_{\mathrm{Borch}}^{\mathrm{vir}}$.
24. Avoid “one-particle Hilbert space” unless qualified.
25. Use “projection-to-E support-locality obstruction,” not generic “locality obstruction.”
26. Use “orientation monodromy character,” not “orientation sign,” when discussing ϵ_o.
27. Keep $D_X = \Delta_5$, $D_5 = 64^{-1}\Delta_5$ consistent.
28. State every imported theorem with its exact role.
29. Separate proof, conjecture, certificate, and open problem typography.
30. Put all variable dictionaries in one table.

The paper’s strongest conceivable form is not a smaller normalization note. It is a **Dirac–Igusa program** with one genuinely new mathematical mechanism:

$$\text{normal-ordered Gram descent from the physical Mukai charge lattice to the Igusa/}$$

Everything should be reorganized around that.

Sources

Sources